The actual forced Navier–Stokes construction in geometric and thermodynamic correspondences

8 September 2026

Abstract

The exact axis data of the released 166-page forced Navier–Stokes construction determine its full first velocity jet. We calculate its five-dimensional Rindler curvature coefficient, boundary stress and momentum flux, and give an invertible representation of its entire force as curved boundary data. We calculate the horizon shear, normal-connection curvature, expansion, and forced area–energy balance with all undetermined matter projections and geometric remainders retained. A stereographic flux map transports the complete forced equation to a round sphere. A specified arithmetic thermometric model gives an exact state, Hamiltonian, and modular time dictionary for the actual vorticity observable. The existence of the input field and the cited derivative constructions are identified source imports. The identities proved here do not assert an exact nonlinear vacuum completion or a terminal spacetime singularity outside the range of the source expansions.

Contents

Conventions and physical units 2

1 Exact jets and force data of the actual constructed velocity 2

1.1 The source fields, with both cutoff sums retained . . . 3
1.2 Why no cutoff correction changes the first axis jet . . . 4
1.3 The full Cartesian tensor and its scalar invariants . . . 5
1.4 The actual axial vorticity balance, with its residual retained . . . 6
1.5 Energy, pressure shifts and the non-gradient force obstruction . . . 7

2 The actual field in the Rindler metric and its momentum constraint 9

2.1 Coordinates, inverses, and every expansion parameter . . . 9
2.2 Brown–York stress, complete momentum, and the matter flux . . . 10
2.3 An explicit curved-boundary representation of the whole force . . . 11
2.4 Coordinate-invariant curvature of the actual axis coefficient . . . 12

3 The actual field in the membrane force, shear, and entropy map 13

3.1 Source, conventions, and the complete parameter map . . . 13
3.2 The geometric coefficient map and its remainders . . . 14
3.3 The force projection, including a discrepancy in the source . . . 15
3.4 The independent null flux and its exact tensor fibres . . . 16
3.5 The actual axis strain and the normal-connection curvature . . . 16
3.6 Forced area and entropy, with the measure and error retained . . . 18
3.7 The separate Rindler background and its temperature . . . 19
3.8 The actual range of the expansion . . . 20

4 The actual forced field on a round sphere and the Schwarzschild boundary constraint 21
4.1 An invertible flux map with the physical coordinates retained 21
4.2 All conformal derivative terms and the exact force 22
4.3 Invariant concentration, dissipation, and a proved pressure obstruction 23
4.4 The source time, the literal metric, and its finite-order domain 24
4.5 The complete global pressure mode and an explicit divergence inverse 26

5 Actual thermal observables, Hamiltonian transport, and modular time 27
5.1 Source fields and the exact axis observable 28
5.2 The retained mechanical-energy and heat balances 29
5.3 The full material flow has an exact Hamiltonian lift 30
5.4 A represented Bost–Connes system with all time conventions 31
5.5 An invertible thermometric image of the actual concentration 33
5.6 The terminal state exists on the represented source algebra 35
5.7 The actual state derivative and all modular clock terms 36

Conventions and physical units

The original Cartesian coordinate x, time t, pressure p_ν, force f_ν, and viscosity ν are preserved. Some source conventions write a field as $u(t,x)$ and others as $u(x,t)$. Whenever time is displayed first in the membrane section it is the exact coordinate permutation $(x,t) \leftrightarrow (t,x)$, with no change to either argument. All new coordinates have a displayed inverse. Three spatial fluid coordinates correspond to a five-dimensional bulk in the cited maps.

The formulas use the original numerical coordinate quantities. To assign laboratory units, specify a length $\ell_f > 0$, a time $t_f > 0$, and a fluid mass density $\rho_f > 0$. The exact unit dictionary is

$$x_{\text{lab}} = \ell_f x, \quad t_{\text{lab}} = t_f t, \quad u_{\text{lab}} = \ell_f t_f^{-1} u_\nu, \quad \nu_{\text{lab}} = \ell_f^2 t_f^{-1} \nu, \quad p_{\text{lab}} = \rho_f \ell_f^2 t_f^{-2} p_\nu, \quad f_{\text{lab}} = \ell_f t_f^{-2} f_\nu.$$

Thus p_ν is kinematic pressure and f_ν is force per unit mass; the inverse multiplies each quantity by the reciprocal factor and evaluates at $x = x_{\text{lab}}/\ell_f, t = t_{\text{lab}}/t_f$. No value of ℓ_f, t_f, ρ_f is selected. A separate gravitational length ℓ_g changes a dimensionless metric g to $ds^2_{\text{grav}} = \ell_g^2 g$. At the unperturbed Rindler boundary its proper time is $d\tau_{\text{proper}} = \ell_g \sqrt{\nu}\, dT/c_{\text{light}}$, whereas the source material time is $dt = \epsilon^2 dT$. These two times are not silently identified. Entropy and thermal formulas below state their gravitational or statistical units explicitly. The additional thermometric energy calibration is specified as part of that model; it is not inferred from the source's pressure.

1 Exact jets and force data of the actual constructed velocity

The input in this section is the actual cutoff-summed and localized field of *Finite Time Blowup for Navier–Stokes*, in the 166-page source edition with SHA-256 `0e779481c4da40bd28d1e642e1d8ca57447d129610df28dfa5a11e9af8ae228f`. The source locations below are printed page numbers. We import its constructed profiles, coefficient axis data, smoothness and support statements, and its residual estimates. This is not an independent verification of the manuscript's entire existence argument. The coordinate, tensor and energy calculations in this section are proved below for those actual fields; no terminal spacetime conclusion is assumed.

1.1 The source fields, with both cutoff sums retained

Write u_*, p_*, f_* for the source's final viscosity-one fields, and retain its fixed choices

$$0 < h < 1/100, \qquad A = \tfrac{1}{2} + h, \qquad D = \tfrac{1}{2} - h, \qquad C > 1, \quad 0 < j_0 \le .05, \quad \Lambda \ge 1, \quad \sigma_* > 0.$$

These are the choices made by the source construction, not independent parameters asserted to produce a solution for every value in those ranges. In particular C retains the source threshold depending on Λ. The right-handed physical cylindrical coordinates are $x_1 = r\cos\theta, x_2 = r\sin\theta, x_3 = z$, and the similarity coordinates in source (4.1), pp. 24–25, are

$$\tau = 1 - t = q(1-\eta^2), \qquad z = q^D\eta, \qquad X = \frac{r^2}{2q}, \qquad d = 1-\eta^2, \qquad L = 1 - 2h\eta^2. \tag{1.1}$$

Here q is the source scalar concentration coordinate. For each $\tau > 0, z \in \mathbb{R}$, it is the unique solution

$$q - z^2 q^{2h} = \tau, \qquad q > |z|^{1/D}, \qquad \eta = zq^{-D}.$$

Indeed the left side is zero at the lower endpoint, tends to infinity, and has derivative $1 - 2hz^2q^{2h-1} = L \ge 1 - 2h > 0$ on this interval. This proves the inverse coordinate map and its smoothness by the implicit derivative, with $|\eta| < 1$. Differentiating the two identities in (1.1) gives

$$q_z = \frac{2\eta q^{1-D}}{L}, \qquad \eta_z = \frac{d}{q^D L}, \qquad q_t = -\frac{1}{L}, \qquad \eta_t = \frac{D\eta}{qL}. \tag{1.2}$$

In particular $q = \tau, \eta = 0, q_z = 0, \eta_z = \tau^{-D}$ on $z = 0$. These are source (4.2), with their constants and time orientation unchanged.

For each coefficient order $n \ge 0$, put $\lambda_n = 2nh$. Source (5.1), p. 46, and (5.27), p. 56, give

$$\begin{aligned} u_{\theta,n} &= \frac{r}{C} q^{-1-h+2nh}\varphi_n(X,\eta), & u_{z,n} &= q^{-A+2nh}U_n(X,\eta), \\ S_n &= q^{1-A+2nh}F_n(X,\eta), & F_n(X,\eta) &= \int_0^X U_n(w,\eta)\,dw, \qquad A_n = \frac{S_n}{r}e_\theta. \end{aligned} \tag{1.3}$$

The symbol F_n in this display is a profile primitive, not a force Taylor coefficient. The realized base uses exactly

$$S = S_0 + \sum_{n\ge1} \chi(c_n q) S_n, \qquad A_{\text{base}} = \frac{S}{r}e_\theta,$$

and the direct azimuthal sum

$$\frac{u_{\theta,B}}{r} = \frac{q^{-1-h}}{C}\left[\varphi_0(X,\eta) + \sum_{n\ge1}\chi(c_n q)q^{2nh}\varphi_n(X,\eta)\right]. \tag{1.4}$$

The cutoffs $\chi(c_n q)$ act on the Stokes streamfunctions before taking curls. The source's further correction stages, including its finite initialization, have the exact representation (9.21), p. 115,

$$\begin{aligned} \mathcal{A} &= \mathcal{A}_0 + \sum_{j\ge1}\chi(a_j q)\mathcal{A}_j, & Be_\theta &= B_0 e_\theta + \sum_{j\ge1}\chi(a_j q)\mathcal{B}_j, \\ p_{\text{loc}} &= p_0 + \sum_{j\ge1}\chi(a_j q)\pi_j, & u_{\text{loc}} &= \nabla\times\mathcal{A} + Be_\theta. \end{aligned} \tag{1.5}$$

The calligraphic representatives here are precisely the source's A_j, B_j, renamed only to distinguish the source exponent A and coefficient streamfunctions. The subscript-zero terms include the realized base and the finite initialization, with the latter's own cutoff as prescribed on p. 114.

The final localization, source (10.4)–(10.5), p. 118, is

$$c(x,t) = \chi_x(x)\chi_t(t), \qquad u_* = \nabla \times (c\mathcal{A}) + cBe_\theta, \qquad p_* = cp_{\text{loc}},$$
$$f_* = \partial_t u_* + (u_* \cdot \nabla)u_* - \Delta u_* + \nabla p_*. \tag{1.6}$$

The fixed spatial cutoff is smooth in (r^2, z), equals one for $r \le r_0/2, |z| \le z_0/2$, and has support compactly contained in $r < r_0, |z| < z_0$. The time cutoff is zero for $1 - t \ge \tau_0$, and one for $0 \le 1 - t \le \tau_0/2$, where $0 < \tau_0 < 1/2$; the source chooses z_0, τ_0 so this support stays inside its local domain. Thus $u_* = cu_{\text{loc}} + \nabla c \times \mathcal{A}$ with the entire localization term retained. Source (10.11), p. 120, extends this same force through time one by its prescribed Taylor coefficients and cutoff sequence. We use its exact preterminal definition above.

For the physical viscosity $\nu > 0$, retain source (10.22), p. 124:

$$y = \frac{x}{\sqrt{\nu}}, \qquad u_\nu(x,t) = \sqrt{\nu}\, u_*(y,t), \quad p_\nu(x,t) = \nu p_*(y,t),$$
$$f_\nu(x,t) = \sqrt{\nu}\, f_*(y,t), \qquad K_\nu = \sqrt{\nu}\, \operatorname{supp} \chi_x. \tag{1.7}$$

The inverse is $x = \sqrt{\nu}\, y$, with multipliers $\nu^{-1/2}, \nu^{-1}, \nu^{-1/2}$ on velocity, pressure and force. Time remains t. The chain rule proves

$$\partial_t u_\nu + (u_\nu \cdot \nabla_x)u_\nu - \nu\Delta_x u_\nu + \nabla_x p_\nu = f_\nu, \qquad \nabla_x \cdot u_\nu = 0.$$

In fact each term on the left is $\sqrt{\nu}$ times the corresponding viscosity-one term. In particular $D_x u_\nu(x,t) = D_y u_*(y,t)$; this equality, rather than a viscosity normalization, will account for the derivative factors below.

1.2 Why no cutoff correction changes the first axis jet

Lemma 5.1, p. 47, gives the exact data

$$\varphi_n(0,\eta) = U_n(0,\eta) = \Pi_n(0,\eta) = 0 \quad (n \ge 1).$$

The extensions and moment corrections of Lemma 5.2 preserve the inner interval. Consequently U_n, φ_n are divisible by X smoothly near zero: for example $U_n(X,\eta) = X \int_0^1 \partial_X U_n(sX,\eta)ds$. Thus F_n in (1.3) is divisible by X^2, including after any fixed η derivative. The radial velocity before the cutoff satisfies $V_n = -\int_0^X Z_{-A+2nh} U_n(w,\eta)dw$, where

$$Z_b U = L^{-1}(2b\eta U + d\partial_\eta U - 2\eta X \partial_X U).$$

The integrand is divisible by X, so V_n is divisible by X^2 as well. Source (5.45), p. 61, retains the extra radial cutoff term explicitly:

$$ru^{\text{cut}}_{r,n} = q^{2nh}\left[\chi(c_n q)V_n - \frac{2\eta}{L}(c_n q)\chi'(c_n q)F_n\right], \qquad u^{\text{cut}}_{z,n} = q^{-A+2nh}\chi(c_n q)U_n. \tag{1.8}$$

At fixed preterminal parameters the two terms in the bracket are $O(r^4)$, hence their radial velocity is $O(r^3)$. The axial correction is $O(r^2)$, and the azimuthal correction is $O(r^3)$. Their first transverse Cartesian derivatives at the axis are therefore zero; their axis values and their axial derivatives of those values are zero as well.

There is no infinite-sum differentiation at a terminal point in this argument. On a neighborhood of each point with $t < 1$, the scalar q has a positive lower bound. Both cutoff sequences tend to infinity and χ has compact support, so only finitely many base orders and correction stages are present. Source Proposition 9.9, pp. 114–117, states that all annular wave and mean representatives, including the finite initialization, vanish on a neighborhood of the axis at each such time. Their local finite sum therefore vanishes on a common neighborhood, including every

Cartesian derivative. Finally, for $0<\tau<\tau_1$, with a fixed source-determined $0<\tau_1\le\tau_0/2$, the localization cutoffs equal one near the origin and every derivative of them vanishes there. This proves that the first axis jet of the actual final field, with both sums and all localization terms retained, is precisely its leading axis jet. It also proves equality of the axis traces on an open (z,t) neighborhood of each of these points, which justifies their axial and time derivatives.

The leading profile's joining and shear modifications preserve its inner profile, as their supports lie beyond a positive radial threshold in the proof of Theorem 4.6, pp. 40–41. The actual leading axis data are explicit. Source (B.1), p. 144, and (B.3), p. 145, followed by (B.12), pp. 146–147, give

$$\begin{gathered} U_0(0,\eta)=4\eta+j_0, \qquad \varphi_0(0,\eta)=\varphi_*(\eta), \\ \varphi_*(\eta)=\exp\left(\Lambda\int_0^\eta \zeta_*(w)\,dw\right), \qquad \zeta_*(\eta)=-\frac{LH_*}{H_*^2+\sigma_*^2}, \qquad H_*=D\eta+d(4\eta+j_0). \end{gathered} \tag{1.9}$$

Here ζ_* is the source auxiliary function. Its full definition, including σ_* and Λ, is retained. In particular

$$\varphi_*(0)=1, \qquad \varphi_*'(0)=-\frac{\Lambda j_0}{j_0^2+\sigma_*^2}. \tag{1.10}$$

The exact viscosity-one axis traces are consequently

$$u_{z,*}(0,0,z,t)=q^{-A}(4\eta+j_0), \qquad \left.\frac{u_{\theta,*}(r,z,t)}{r}\right|_{r=0}=\frac{1}{C}q^{-1-h}\varphi_*(\eta). \tag{1.11}$$

The quotient denotes its smooth extension; no division by zero is performed. Formula (1.11) is valid where the preceding localization and local-domain conditions hold.

1.3 The full Cartesian tensor and its scalar invariants

Set $G_{ij}=\partial_{x_j}(u_\nu)_i$, with the indicated order of the indices. In the axis neighborhood, smoothness and rotational symmetry give the actual Cartesian representation

$$(u_\nu)_1=a(r^2,z,t)x_1-b(r^2,z,t)x_2, \quad (u_\nu)_2=b(r^2,z,t)x_1+a(r^2,z,t)x_2, \quad (u_\nu)_3=w(r^2,z,t).$$

For the source field this representation also follows directly from (4.5), p. 25, the smooth coefficient factors just established and its axisymmetric base potentials. At the axis the transverse derivatives of w vanish, and the axial derivatives of the first two components vanish. Incompressibility at the axis gives $2a+\partial_z w=0$. Equations (1.2) and (1.11) give

$$\partial_z[q^{-A}(4\eta+j_0)]\Big|_{z=0}=4\tau^{-A-D}=\frac{4}{\tau}.$$

Under (1.7) the one spatial derivative cancels the velocity multiplier. It follows that, at the fixed physical origin for every $0<\tau<\tau_1$,

$$\boxed{u_\nu(0,1-\tau)=\sqrt{\nu}\,j_0\tau^{-A}e_3, \qquad G(0,1-\tau)=\begin{pmatrix}-2/\tau & -b_\tau & 0\\ b_\tau & -2/\tau & 0\\ 0 & 0 & 4/\tau\end{pmatrix}, \qquad b_\tau=\frac{1}{C}\tau^{-1-h}.} \tag{1.12}$$

In particular the positive source datum j_0 gives an exact fixed-origin velocity growth as well as the source's separate nonzero-radius swirl sample. No source remainder at that separate sample has been removed by this axis calculation.

Use the physical Euclidean metric δ_{ij} and orientation $dx_1 \wedge dx_2 \wedge dx_3$. Define $S = (G+G^\mathsf{T})/2$, $\Omega = (G - G^\mathsf{T})/2$ and $\omega = \nabla \times u_\nu$. Taking the indicated matrix sum, difference and curl proves

$$S = \operatorname{diag}(-2/\tau, -2/\tau, 4/\tau), \qquad \Omega = \begin{pmatrix} 0 & -b_\tau & 0 \\ b_\tau & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad \omega = 2b_\tau e_3, \tag{1.13}$$

$$\begin{aligned} &\operatorname{tr} S = 0, && S : S = \frac{24}{\tau^2}, && \Omega : \Omega = \frac{2}{C^2}\tau^{-2-2h}, \\ &|\omega|^2 = \frac{4}{C^2}\tau^{-2-2h}, && G : G = \frac{24}{\tau^2} + \frac{2}{C^2}\tau^{-2-2h}, && (2S) : (2S) = \frac{96}{\tau^2}. \end{aligned} \tag{1.14}$$

Here $M : N = \sum_{i,j} M_{ij}N_{ij}$; thus these are actual scalar contractions with the physical spatial metric. For example, under an orthonormal basis change $G \mapsto OGO^\mathsf{T}$, cyclicity of trace proves $G : G = \operatorname{tr}(GG^\mathsf{T})$ is unchanged. Under general spatial coordinates the same values are obtained by covariantly differentiating the Euclidean velocity and contracting with the transformed Euclidean metric, because both operations are tensorial. They are spatial invariants of this fluid data; a spacetime curvature invariant requires its additional metric map.

The remaining elementary matrix invariants, useful for retaining rotation as well as strain, are

$$\begin{aligned} \operatorname{tr}(G^2) &= \frac{24}{\tau^2} - \frac{2}{C^2}\tau^{-2-2h}, \\ \det G &= \frac{4}{\tau}\left(\frac{4}{\tau^2} + \frac{1}{C^2}\tau^{-2-2h}\right), \qquad S\omega = \frac{4}{\tau}\,\omega. \end{aligned} \tag{1.15}$$

The two-by-two block squares to a block with diagonal $4/\tau^2 - b_\tau^2$, proving the trace; its determinant is $4/\tau^2 + b_\tau^2$, proving the determinant formula. The stretching identity follows by multiplying S by $2b_\tau e_3$. The pointwise viscous dissipation per unit mass of an incompressible Newtonian fluid, defined by $2\nu S : S$, is therefore exactly

$$2\nu S : S = \frac{48\nu}{\tau^2} \quad \text{at } x = 0. \tag{1.16}$$

1.4 The actual axial vorticity balance, with its residual retained

The axis curl is $2(u_\theta/r)e_3$, by the Cartesian representation above. Hence (1.11), the coordinate derivatives and the physical map give

$$\begin{aligned} \partial_t \omega_z(0, 1-\tau) &= \frac{2}{C}(1+h)\tau^{-2-h}, \\ \partial_{x_3}\omega_z(0, 1-\tau) &= \frac{2}{C\sqrt{\nu}}\tau^{-1-h-D}\varphi_*'(0) = \frac{2}{C\sqrt{\nu}}\tau^{-3/2}\varphi_*'(0). \end{aligned} \tag{1.17}$$

The first derivative is taken at the fixed origin. The material derivative also includes the actual nonzero axial velocity in (1.12); the origin is not declared a fixed material particle. Since $A + D = 1$, its advective term is

$$(u_\nu \cdot \nabla)\omega_z(0, 1-\tau) = \frac{2}{C} j_0 \varphi_*'(0)\tau^{-2-h}.$$

For completeness, the vector identity $(u \cdot \nabla)u = \nabla(|u|^2/2) - u \times (\nabla \times u)$ follows by expanding the two Levi-Civita symbols in each component. Its curl, using $\nabla \cdot u = 0$ and $\nabla \cdot (\nabla \times u) = 0$, gives

$$\partial_t \omega + (u_\nu \cdot \nabla)\omega - (\omega \cdot \nabla)u_\nu = \nu\Delta\omega + \nabla \times f_\nu.$$

In the axial component the stretching term is $(\omega \cdot \nabla)(u_\nu)_3 = 8C^{-1}\tau^{-2-h}$. Substituting the three exact terms and (1.10) proves

$$\boxed{\nu \Delta \omega_z(0, 1-\tau) + (\nabla \times f_\nu)_z(0, 1-\tau) = \frac{2}{C}\left[h - 3 - \frac{\Lambda j_0^2}{j_0^2 + \sigma_*^2}\right]\tau^{-2-h}.} \tag{1.18}$$

This retains the actual force. More explicitly, define $r_{\nu,\omega}(\tau) = (\nabla \times f_\nu)_z(0, 1-\tau)$. Source (10.9), p. 119, proves that for every $N > 0$ there are finite source constants C_N and $\tau_N > 0$ with $|r_{\nu,\omega}(\tau)| \le C_N \tau^N$ for $0 < \tau < \tau_N$. The cancellation of $\sqrt{\nu}$ against one spatial derivative proves $r_{\nu,\omega} = r_{*,\omega}$ exactly. Thus the full Laplacian is

$$\Delta \omega_z(0, 1-\tau) = \frac{2}{\nu C}\left[h - 3 - \frac{\Lambda j_0^2}{j_0^2 + \sigma_*^2}\right]\tau^{-2-h} - \frac{1}{\nu} r_{*,\omega}(\tau).$$

The bracket is strictly negative, since $h < 1/100$, $\Lambda > 0$, $j_0 > 0$ and $\sigma_* > 0$. This computes a genuine derivative of the actual vorticity, with its original residual, without replacing the forced equation by a homogeneous one.

1.5 Energy, pressure shifts and the non-gradient force obstruction

For every $T < 1$, the actual u_ν, p_ν are smooth with fixed compact spatial support K_ν, and $u_\nu(\cdot, 0) = 0$. Define $E_\nu(t) = \frac{1}{2}\|u_\nu(t)\|_2^2$. Multiply the exact momentum equation by u_ν, sum components and integrate on $\mathbb{R}^3$. The time term is E_ν'. The convection term is the integral of $\nabla \cdot (u_\nu |u_\nu|^2/2)$, hence zero by compact support. The pressure term is the integral of $\nabla \cdot (p_\nu u_\nu) - p_\nu \nabla \cdot u_\nu$, also zero. One integration by parts gives $-\int u_\nu \cdot \Delta u_\nu = \|\nabla u_\nu\|_2^2$. Consequently the exact identity is

$$E_\nu(T) + \nu \int_0^T \|\nabla u_\nu(t)\|_2^2 dt = \int_0^T \int_{\mathbb{R}^3} u_\nu \cdot f_\nu \, dx \, dt. \tag{1.19}$$

The source smooth force extension gives $F_\nu(T) = \int_0^T \|f_\nu(t)\|_2 dt < \infty$, including at $T = 1$. To retain the full estimate, divide the differential energy identity by $(\|u_\nu\|_2^2 + \delta^2)^{1/2}$, discard its nonnegative dissipation and use Cauchy–Schwarz. The left time term is the derivative of that square root. Integration and $\delta \downarrow 0$ give $\|u_\nu(t)\|_2 \le F_\nu(t)$. Substitution into (1.19) proves

$$\|u_\nu(T)\|_2^2 + 2\nu \int_0^T \|\nabla u_\nu(t)\|_2^2 dt \le F_\nu(T)^2, \qquad F_\nu(1) = \nu^{5/4} F_*(1). \tag{1.20}$$

The last factor follows from the $\sqrt{\nu}$ force multiplier and $dx = \nu^{3/2} dy$. The same change of variables gives $\|u_\nu\|_2^2 = \nu^{5/2}\|u_*\|_2^2$ and the identical $\nu^{5/2}$ factor for the integrated viscous dissipation. Moreover

$$\int_{\mathbb{R}^3} \partial_j (u_\nu)_i \, \partial_i (u_\nu)_j \, dx = -\int_{\mathbb{R}^3} (u_\nu)_i \partial_i \partial_j (u_\nu)_j \, dx = 0,$$

so $\int 2S : S = \int |\nabla u_\nu|^2$ exactly. Thus the divergence of the pointwise quantity (1.16) is compatible with the finite full spacetime integral in (1.20); a point evaluation is not that integral.

Here is the precise pressure transformation that removes exactly a gradient force. For any retained smooth scalar $\psi(x, t)$, set

$$\widetilde{p} = p_\nu - \psi, \qquad \widetilde{f} = f_\nu - \nabla \psi. \tag{1.21}$$

Then the same actual velocity satisfies the same equation with $(\widetilde{p}, \widetilde{f})$, by subtracting $\nabla \psi$ from its two sides. The inverse, with ψ retained as part of the data, is $p_\nu = \widetilde{p} + \psi$, $f_\nu = \widetilde{f} + \nabla \psi$. In particular this map removes the entire force exactly when $f_\nu = \nabla \psi$.

For the actual input that last equality cannot hold on any whole interval $[0,T]$ with $1-\tau_1 < T < 1$. Indeed a gradient force has zero work against the compactly supported divergence-free u_ν, by integration by parts. Equation (1.19) would give $E_\nu(T) = 0$. But (1.12) gives $(u_\nu)_3(0,T) = \sqrt{\nu} j_0(1-T)^{-A} > 0$; continuity implies that its square has strictly positive integral on a ball. Thus $E_\nu(T) > 0$, a contradiction. This proves more than a nonzero force: the force has a nontrivial class modulo smooth gradient forces on every such initial interval.

It also proves that $\nabla \times f_\nu$ is nonzero at some space-time point of each of those initial intervals. To prove this without a topological assumption left implicit, if a smooth force on $\mathbb{R}^3$ has zero curl, put

$$\psi(x,t) = \int_0^1 x_j (f_\nu)_j(sx,t)\, ds \quad \text{(summation over } j = 1,2,3).$$

The curl-free identity $\partial_k f_j = \partial_j f_k$ gives

$$\partial_k \psi = \int_0^1 \left[f_k(sx,t) + s x_j \partial_j f_k(sx,t) \right] ds = \int_0^1 \frac{d}{ds}[s f_k(sx,t)] ds = f_k(x,t).$$

This potential is smooth jointly in space and time, so zero curl on the whole initial interval would give the excluded gradient force. The argument does not assert nonzero curl at every late time, or at the concentrating origin: the actual force derivatives at that origin have the flat estimates used in (1.18).

One can specify the surviving force component by an exact integral map. Let $\Gamma(x) = -1/(4\pi|x|)$, the fundamental solution of the spatial Laplacian, and define

$$\psi_\nu = \Gamma * (\nabla \cdot f_\nu), \qquad g_\nu = f_\nu - \nabla \psi_\nu, \qquad \widehat{p}_\nu = p_\nu - \psi_\nu. \tag{1.22}$$

The force is smooth with compact support, so the convolution is defined by a locally integrable kernel and is smooth. More explicitly, every derivative can be moved to the compactly supported smooth factor; the integral is then absolutely convergent near the kernel singularity and locally uniform in the free variables. The identity $\Delta\Gamma = \delta_0$ follows by integrating by parts on the complement of a radius-ϵ ball: the derivative boundary term tends to the test function value at zero because $4\pi\epsilon^2 \partial_r \Gamma = 1$, and the boundary term with the test-function derivative tends to zero. Therefore

$$\nabla \cdot g_\nu = 0, \qquad \nabla \times g_\nu = \nabla \times f_\nu, \qquad \partial_t u_\nu + (u_\nu \cdot \nabla) u_\nu - \nu \Delta u_\nu + \nabla \widehat{p}_\nu = g_\nu.$$

This is an explicit source force-to-pressure decomposition, with inverse $(g_\nu, \psi_\nu, \widehat{p}_\nu) \mapsto (g_\nu + \nabla \psi_\nu, \widehat{p}_\nu + \psi_\nu)$. Its image is the set of triples satisfying the convolution relation in (1.22); neither a new arbitrary scalar nor a new boundary condition is substituted for the source pressure. The transformed fields $g_\nu, \widehat{p}_\nu$ need not have compact support, while the original f_ν, p_ν retain their specified support and their displayed inverse. This map also retains the source viscosity scaling. Substitute $x = \sqrt{\nu} y$ and $x' = \sqrt{\nu} y'$ in the defining convolution. Its kernel contributes $\nu^{-1/2}$, the volume element contributes $\nu^{3/2}$, and the divergence of the rescaled force contributes one. Hence

$$\psi_\nu(x,t) = \nu \psi_*(y,t), \qquad g_\nu(x,t) = \sqrt{\nu} g_*(y,t), \qquad \widehat{p}_\nu(x,t) = \nu \widehat{p}_*(y,t).$$

Since u_ν has compact support,

$$\int_0^T \int u_\nu \cdot g_\nu = \int_0^T \int u_\nu \cdot f_\nu = E_\nu(T) + \nu \int_0^T \|\nabla u_\nu\|_2^2 dt > 0 \quad (1 - \tau_1 < T < 1).$$

Thus g_ν is the explicitly retained, nonzero force component that any geometric, boundary or matter realization must account for. The conclusion is a proved obstruction to the pressure-only removal of this actual force, together with the exact residual force map.

2 The actual field in the Rindler metric and its momentum constraint

All statements concerning the existence of the input refer to the precise 166-page released construction [1]. Its existence proof is source-imported. The tensor identities below are proved for that specified input; they do not assert independent certification of its all-order construction. Let $u_\nu(x,t), p_\nu(x,t), f_\nu(x,t)$ be exactly the fields defined in Section 1, with $\nu > 0$ and $0 \le t < 1$. In particular their pressure and force are retained, including every spatial and temporal cutoff derivative. The bulk dimension in the present problem is five: three fluid spatial coordinates, one fluid time, and one radial coordinate. No four-dimensional Einstein assertion follows from this three-dimensional fluid correspondence.

2.1 Coordinates, inverses, and every expansion parameter

Introduce a positive parameter ϵ, new coordinates (Y,T), and the exact diffeomorphism

$$x = \epsilon Y, \quad t = \epsilon^2 T, \qquad Y = x/\epsilon, \quad T = t/\epsilon^2. \tag{2.1}$$

The transported fields are

$$\begin{aligned} v(Y,T) &= \epsilon u_\nu(\epsilon Y, \epsilon^2 T), & P(Y,T) &= \epsilon^2 p_\nu(\epsilon Y, \epsilon^2 T), \\ F(Y,T) &= \epsilon^3 f_\nu(\epsilon Y, \epsilon^2 T), & r_c &= \nu. \end{aligned} \tag{2.2}$$

Their inverses are $u_\nu(x,t) = \epsilon^{-1} v(x/\epsilon, t/\epsilon^2)$, $p_\nu = \epsilon^{-2} P(x/\epsilon, t/\epsilon^2)$, and $f_\nu = \epsilon^{-3} F(x/\epsilon, t/\epsilon^2)$. The viscosity has not changed. Chain differentiation, term by term, gives

$$\partial_T v_i + v^j \partial_j v_i - \nu \partial^2 v_i + \partial_i P = F_i, \qquad \partial_i v^i = 0. \tag{2.3}$$

Each momentum term acquires exactly ϵ^3, and the divergence acquires ϵ^2. The spatial support is $\epsilon^{-1} K_\nu$; zero initial velocity and the specified pressure support persist.

We use r only for the gravitational radial coordinate in this section. The source cylindrical radius, when needed, is denoted $r_{\rm cyl}$. On $0 \le r \le \nu$ the metric displayed in BKLS [2], equation (14), is the following tensor with its omitted terms retained as a named remainder:

$$\begin{aligned} g_{[3]} = &- r\, dT^2 + 2\, dT\, dr + \delta_{ij} dY^i dY^j \\ &- 2(1 - r/\nu) v_i\, dY^i dT - 2 v_i \nu^{-1} dY^i dr \\ &+ (1 - r/\nu) \Big[(v^2 + 2P) dT^2 + \nu^{-1} v_i v_j dY^i dY^j \Big] + (v^2 + 2P) \nu^{-1} dT dr \\ &- (r^2 - \nu^2) \nu^{-1} \partial^2 v_i\, dY^i dT, \qquad g = g_{[3]} + \mathcal{E}_{\ge 4}. \end{aligned} \tag{2.4}$$

Products of differentials denote the symmetric line-element convention; thus the coefficient $-2v_i/\nu$ is twice g_{ri}. The four displayed lines have hydrodynamic weights zero, one, two, and three, where v, ∂_i have weight one and P, ∂_T weight two. Here $\mathcal{E}_{\ge 4}$ denotes the undisplayed formal metric terms; it is not an asserted convergent vacuum correction for the forced field. The finite tensor $g_{[3]}$ itself is explicitly defined and smooth on every closed subterminal interval. For sufficiently small ϵ on such an interval and on a compact radial range, it is Lorentzian by continuity from the nondegenerate Rindler metric. All coefficients and derivatives of the source fields needed for this assertion are bounded there. The resulting smallness constants may depend on the interval.

At $r = \nu$, $dr = 0$, so every displayed perturbation in the induced metric vanishes and $\gamma = -\nu dT^2 + dY^2$. The inverse of the displayed metric map at hydrodynamic weights one and two is explicit: its ri component gives $v_i = -\nu (g_{ri})_{[1]}$, and at any $r \ne \nu$ its TT coefficient gives $P = \frac{1}{2}[(g_{TT} + r)_{[2]}/(1 - r/\nu) - v^2]$. The subscripts mean coefficient extraction in the stated grading, not an inverse of unknown all-order Einstein solutions.

BKLS's alternate coordinates are also retained with their inverse:

$$\widehat{Y} = \epsilon Y/\nu = x/\nu, \quad \widehat{T} = \epsilon^2 T/\nu = t/\nu, \quad \widehat{r} = r/\nu, \quad \lambda = \epsilon^2/\nu; \quad (Y, T, r) = (\nu\widehat{Y}/\epsilon, \nu\widehat{T}/\epsilon^2, \nu\widehat{r}). \tag{2.5}$$

The fields there are $\widehat{v}(\widehat{Y}, \widehat{T}) = u_\nu(\nu\widehat{Y}, \nu\widehat{T})$ and $\widehat{P} = p_\nu(\nu\widehat{Y}, \nu\widehat{T})$. Their viscosity-one equation has force $\widehat{F} = \nu f_\nu(\nu\widehat{Y}, \nu\widehat{T})$, since every momentum derivative acquires ν. This equation is an image with the explicit inverse $t = \nu\widehat{T}, x = \nu\widehat{Y}$; the original viscosity and time remain as in (2.2). The additional constant metric homothety is $\widehat{g} = \epsilon^2\nu^{-2}g$, with inverse $g = \nu^2\epsilon^{-2}\widehat{g}$. For a covariant metric multiplied by $c > 0$, its connection and mixed curvature do not change, its covariant curvature gains c, and four inverse metrics in the full contraction give $\mathrm{Riem}^2(cg) = c^{-2}\,\mathrm{Riem}^2(g)$. This proves the curvature scaling without identifying a metric homothety with a coordinate transformation.

2.2 Brown–York stress, complete momentum, and the matter flux

Adopt BKLS's outward normal and sign $K_{ab} = \frac{1}{2}\mathcal{L}_N\gamma_{ab}$, and keep its numerical stress $\mathsf{T}_{ab} = 2(\gamma_{ab}K - K_{ab})$. Restoring Newton's constant G_N gives the physical Brown–York stress $\mathsf{T}^{\mathrm{phys}}_{ab} = \mathsf{T}_{ab}/(16\pi G_N)$. Through the source's displayed order,

$$\begin{aligned}
N &= \nu^{-1/2}\partial_T + \nu^{1/2}(1 - P/\nu)\partial_r + \nu^{-1/2}v^i\partial_i + \mathcal{N}_{\geq 3},\\
\mathsf{T}_{TT} &= \nu^{-1/2}v^2 + \mathcal{B}_{TT,\geq 3}, \qquad\qquad \mathsf{T}_{Ti} = -\nu^{-1/2}v_i + \mathcal{B}_{Ti,\geq 3},\\
\mathsf{T}_{ij} &= \nu^{-1/2}\delta_{ij} + \nu^{-3/2}(v_iv_j + P\delta_{ij}) - \nu^{-1/2}(\partial_i v_j + \partial_j v_i) + \mathcal{B}_{ij,\geq 3}.
\end{aligned} \tag{2.6}$$

These are BKLS equations (17)–(18), including its symmetric tensor interpretation; replacing the last symmetric expression by one unsymmetrized tensor would be incorrect. They can also be recovered by inserting (2.4) in $N = g^{r\mu}\partial_\mu/\sqrt{g^{rr}}$ and $K_{ab} = \frac{1}{2}N^\mu\partial_\mu g_{ab} + \frac{1}{2}(g_{\mu b}\partial_a N^\mu + g_{a\mu}\partial_b N^\mu)$ at $r = \nu$. The definition gives the algebraic inverse $K = \gamma^{ab}\mathsf{T}_{ab}/6$ and $K_{ab} = \gamma_{ab}\gamma^{cd}\mathsf{T}_{cd}/6 - \mathsf{T}_{ab}/2$.

There is a direct check of every momentum term. With $\gamma^{TT} = -1/\nu$, $\gamma^{ij} = \delta^{ij}$, differentiation of the displayed coefficients gives

$$\begin{aligned}
\nu^{3/2}(\partial^a\mathsf{T}_{ai})_{[3]} &= \partial_T v_i + \partial_j(v_jv_i) + \partial_i P - \nu\partial_j(\partial_j v_i + \partial_i v_j)\\
&= \partial_T v_i + v^j\partial_j v_i - \nu\partial^2 v_i + \partial_i P = F_i.
\end{aligned} \tag{2.7}$$

The two removed terms are explicitly $v_i\partial_j v_j$ and $-\nu\partial_i\partial_j v_j$, both zero by incompressibility. The temporal constraint has weight-two coefficient $\nu^{3/2}(\partial_a\mathsf{T}^{aT})_{[2]} = \partial_i v^i = 0$. Thus the actual smooth forcing is exactly the first nonzero momentum constraint defect of the intrinsically flat-boundary ansatz.

For completeness fix the curvature convention $R^a{}_{bcd} = \partial_c\Gamma^a_{db} - \partial_d\Gamma^a_{cb} + \Gamma^a_{ce}\Gamma^e_{db} - \Gamma^a_{de}\Gamma^e_{cb}$. In Gaussian normal coordinates $g = dn^2 + \gamma_{ab}dy^ady^b$, $\Gamma^n_{ab} = -K_{ab}$ and $\Gamma^a_{nb} = K^a{}_b$. Substitution in the Ricci formula gives $R_{nb} = D_aK^a{}_b - D_bK$; this proves Codazzi with the chosen sign. Consequently, in every coordinate system,

$$D_a\mathsf{T}^a{}_b = -2G(N, e_b), \qquad \big(G(N, e_i)\big)_{[3]} = -\frac{F_i}{2\nu^{3/2}}, \qquad \big(T^{\mathrm{matter}}(N, e_i)\big)_{[3]} = -\frac{F_i}{16\pi G_N\nu^{3/2}}. \tag{2.8}$$

The last equality uses $G_{\mu\nu} = 8\pi G_N T^{\mathrm{matter}}_{\mu\nu}$. It specifies the required normal momentum flux, not a full matter model or a proof of an energy condition. For the actual finite Lorentzian $g_{[3]}$, the tensor $G[g_{[3]}]/(8\pi G_N)$ is an exact, conserved geometrically defined effective stress, by the contracted Bianchi identity; its higher coefficients are not suppressed in this definition. This is an exact metric-to-stress map, whose prescribed leading flux is (2.8). It does not change the original force into zero.

2.3 An explicit curved-boundary representation of the whole force

The absence of vacuum momentum conservation for the fixed flat boundary does not exclude vacuum boundary driving. Here is a constructive boundary map for every component of the actual force, including its gradient part. Work first in the original (x,t) variables. Denote the material diffeomorphism by X_t, $\dot X_t(q) = u_\nu(X_t(q),t)$, $X_0(q)=q$. On each $[0,t_0]$, $t_0<1$, this is a smooth diffeomorphism: bounded smooth compact support gives the usual Picard iteration on successive bounded intervals; the inverse is obtained by the backwards ODE, and $\partial_t \det DX_t = (\operatorname{div} u_\nu)\det DX_t = 0$.

Regard $f_\nu = f_{\nu,i}dx^i$ as the original force one-form, and define

$$A(t) = -(X_t^{-1})^* \int_0^t X_s^*(f_\nu(s))\,ds, \qquad \Phi(t,x) = -u_\nu^i(t,x)A_i(t,x). \tag{2.9}$$

The pullback of a one-form is explicitly $(X_s^*f)_j(q) = f_i(X_s(q),s)\partial_j X_s^i(q)$; the inverse pullback uses the inverse matrix DX_t^{-1}. Differentiating this component expression proves $\partial_t(X_t^*A) = X_t^*(\partial_t A + \mathcal{L}_{u_\nu}A)$, where $(\mathcal{L}_uA)_i = u^j\partial_jA_i + A_j\partial_iu^j$. Equation (2.9) therefore solves, uniquely,

$$(\partial_t + \mathcal{L}_{u_\nu})A = -f_\nu, \quad A(0)=0; \qquad f_\nu = -(\partial_t+\mathcal{L}_{u_\nu})A. \tag{2.10}$$

Uniqueness follows by pulling a homogeneous solution back by X_t. The displayed differential formula is the inverse for this specified initial condition. Since X_t is the identity off K_ν and the force vanishes there, A and Φ are smooth and supported in K_ν on every subterminal interval. No inverse Laplacian or loss of a force component enters this map.

Scale $a_i(Y,T) = \epsilon A_i(\epsilon Y, \epsilon^2 T)$ and $\phi(Y,T) = \epsilon^2\Phi(\epsilon Y,\epsilon^2 T)$. Then $\phi = -v^ia_i$ and the exact scaled identity is $\partial_T a + \mathcal{L}_v a = -F$. Define a boundary metric

$$\gamma_{a,\phi} = -\nu dT^2 + \delta_{ij}dY^idY^j + 2a_i\,dTdY^i + 2\phi\,dT^2. \tag{2.11}$$

It has lapse square $N_{\rm ADM}^2 = \nu - 2\phi + a^2 = \nu - v^2 + |a+v|^2$; its determinant is $-N_{\rm ADM}^2$. On $v^2<\nu$ it is Lorentzian and $U = (\partial_T + v^i\partial_i)/\sqrt{\nu - v^2}$ is timelike unit, because $\gamma_{a,\phi}(\partial_T+v,\partial_T+v) = -\nu+v^2$. Its inverse has $\gamma^{TT} = -N_{\rm ADM}^{-2}$, $\gamma^{Ti} = a_iN_{\rm ADM}^{-2}$, and $\gamma^{ij} = \delta^{ij} - a_ia_jN_{\rm ADM}^{-2}$. This is an exact timelike domain for the constructed boundary data.

The forcing encoded by this metric is verified directly, without an assumed nonlinear bulk solution. For a timelike curve with coordinate velocity w, put $N_w^2 = \nu - w^2 - 2a\cdot w - 2\phi$. The nonrelativistic term in its proper-time action is proportional to $L = \frac12 w^2 + a\cdot w + \phi$; alternatively use $U^b\nabla_bU_i = N_w^{-1}(\partial_T + w\cdot\nabla)[(w_i+a_i)/N_w] - \frac12 U^bU^c\partial_i\gamma_{bc}$. Substituting $w=v$, expanding only by the stated weights, and retaining the next terms as a remainder gives

$$U^b\nabla_bU_i = \nu^{-1}\Big[\partial_Tv_i + v^j\partial_jv_i + \partial_Ta_i + v^j(\partial_ja_i - \partial_ia_j) - \partial_i\phi\Big] + \mathcal{A}_{i,\geq 5}. \tag{2.12}$$

Indeed $N_w = \sqrt{\nu} + O(\epsilon^2)$, $\partial_ia_j = O(\epsilon^2)$, and $(\partial_T + v\cdot\nabla)N_w = O(\epsilon^4)$, so the terms omitted in this displayed acceleration have weight at least five. The exact acceleration remains the preceding unexpanded expression. The symmetric spatial projected velocity derivative is $\sigma_{ij} = \nu^{-1/2}(\partial_iv_j + \partial_jv_i)/2 + \mathcal{S}_{ij,\geq 4}$: the symmetric derivative of a in U_i cancels the Christoffel term, since $\Gamma^T_{ij} = -(\partial_ia_j+\partial_ja_i)/(2\nu) + O(\epsilon^4)$ and $U_T = -\sqrt{\nu} + O(\epsilon^2)$. Here is the full constitutive tensor used in this boundary calculation. With all contractions made using $\gamma_{a,\phi}$, define

$$H_{ab} = \gamma_{ab} + U_aU_b, \qquad \vartheta = \nabla_aU^a, \qquad \sigma_{ab} = H_a{}^cH_b{}^d\nabla_{(c}U_{d)} - \tfrac13\vartheta H_{ab},$$
$$p_{\rm rel} = \nu^{-1/2} + P\nu^{-3/2}, \qquad \mathsf{T}^{\rm bdry}_{ab} = p_{\rm rel}H_{ab} - 2\sigma_{ab}.$$

This specifies zero rest energy and shear coefficient 1 in BKLS stress units. It is an explicitly chosen boundary constitutive tensor, not a claim that the whole tensor is already realized as the extrinsic curvature of a nonlinear vacuum bulk. Its spatial divergence is $\partial_ip_{\rm rel} + U_iU^a\partial_ap_{\rm rel} +$

$p_{\rm rel}(\vartheta U_i + U^a\nabla_a U_i) - 2\nabla_a \sigma^a{}_i$. The second and third terms involving U_i have weight at least five: $U^a\partial_a P$ has weight four and ϑ has weight four by incompressibility. The leading dissipative divergence is $\nu^{-1/2}\partial^2 v_i$, since $\partial_j(\partial_j v_i + \partial_i v_j) = \partial^2 v_i$ by $\partial_j v^j = 0$; its other projection and connection contributions start at weight five. This proves that the spatial conservation equation $\nabla_a \mathsf{T}^a{}_i = 0$ has weight-three part

$$\partial_T v_i + v^j\partial_j v_i - \nu\partial^2 v_i + \partial_i P = -\partial_T a_i + v^j(\partial_i a_j - \partial_j a_i) + \partial_i\phi = F_i. \tag{2.13}$$

The last equality is exact: its middle expression equals $-\partial_T a_i - (\mathcal{L}_v a)_i + \partial_i(v\cdot a + \phi)$, and $v\cdot a + \phi = 0$. Thus (2.9) represents the actual force as geometry in the hydrodynamic boundary equation, with all terms retained. The exact Lorentzian metric, inverse map, and explicit acceleration have been constructed. A vacuum bulk completion of this nonlinear driven boundary data is not supplied by BKLS Section 5.2, which constructs a linear impulsive forcing example and expressly does not construct its nonlinear bulk analogue. We do not assign the uncomputed bulk to it.

2.4 Coordinate-invariant curvature of the actual axis coefficient

The squared Riemann tensor is a Lorentzian scalar; it need not be positive. BKLS Section 5.3 gives its weight-four coefficient

$$(\mathrm{Riem}^2(g_{[3]}))_{[4]} = -\frac{3}{2\nu^2}(\partial_i v_j - \partial_j v_i)(\partial_i v_j - \partial_j v_i). \tag{2.14}$$

The following local calculation proves this coefficient for the actual axis jet independently of the unforced equation. Set a formal first-jet matrix $M = \begin{pmatrix} a & -b & 0 \\ b & a & 0 \\ 0 & 0 & c \end{pmatrix}$, and take the weight-two part of the linear metric perturbation $h_{Ti} = -(1 - r/\nu)(MY)_i$, $h_{ri} = -(MY)_i/\nu$. The nonzero background connection is $\Gamma^T_{TT} = 1/2$, $\Gamma^r_{TT} = r/2$, $\Gamma^r_{Tr} = \Gamma^r_{rT} = -1/2$. With $g_0 = -rdT^2 + 2dTdr + dY^2$, the full coefficient is obtained by

$$\begin{aligned}
\delta g^{-1} &= -g_0^{-1} h g_0^{-1}, \\
\delta\Gamma^A_{BC} &= \tfrac{1}{2} g_0^{AD}(\partial_B h_{DC} + \partial_C h_{DB} - \partial_D h_{BC}) + \tfrac{1}{2}\delta g^{AD}(\partial_B (g_0)_{DC} + \partial_C (g_0)_{DB} - \partial_D (g_0)_{BC}), \\
\delta R^A{}_{BCD} &= \partial_C \delta\Gamma^A_{DB} - \partial_D \delta\Gamma^A_{CB} + \Gamma^A_{CE}\delta\Gamma^E_{DB} + \delta\Gamma^A_{CE}\Gamma^E_{DB} - \Gamma^A_{DE}\delta\Gamma^E_{CB} - \delta\Gamma^A_{DE}\Gamma^E_{CB}.
\end{aligned}$$

For a shorter component verification, let $W_{ij} = \partial_i v_j - \partial_j v_i$ at this order. The independent nonzero lower curvature components from these formulas are

$$R^{[2]}_{Trij} = \frac{W_{ij}}{2\nu}, \qquad R^{[2]}_{Tirj} = \frac{W_{ij}}{4\nu}, \qquad R^{[2]}_{TiTj} = \frac{\partial_i v_j + \partial_j v_i}{4}, \qquad R^{[2]}_{rirj} = 0. \tag{2.15}$$

All other components follow by curvature symmetries, or are zero. In particular the symmetric-gradient components have zero square and zero cross-contraction with these antisymmetric ones: raising a T index uses $g_0^{Tr} = 1$, while $g_0^{TT} = 0$ and $R^{[2]}_{rirj} = 0$. Contracting the antisymmetric components yields $-3\sum_{ij} W_{ij}^2/(2\nu^2) = -12b^2/\nu^2$. The exact symbolic record accompanying this manuscript enumerates every nonzero component and repeats that contraction using rational arithmetic. Algebraic terms involving the original zero-jet velocity and pressure are retained in (2.4); they give zero curvature at weight two, as follows by the same substitution with constant v, P. Terms mixing a zero-jet velocity with a first spatial derivative have weight three. Thus the calculation extracts a coefficient of the original field and does not replace it by the linear test jet.

At the actual axis, Section 1 proves $b = \epsilon^2 C^{-1}\tau^{-1-h}$, $a = -2\epsilon^2/\tau$, $c = 4\epsilon^2/\tau$, at $Y = 0$, $T = (1-\tau)/\epsilon^2$. Consequently

$$(\mathrm{Riem}^2)_{[4]} = -\frac{12\epsilon^4}{\nu^2 C^2}\tau^{-2-2h}. \tag{2.16}$$

For the finite tensor $g_{[3]}$, retain the exact scalar remainder

$$\mathcal{K}_{\mathrm{rem}} = \mathrm{Riem}^2(g_{[3]}) + 12\epsilon^4(\nu^2 C^2)^{-1}\tau^{-2-2h}.$$

On every fixed smooth subterminal compact set it has strictly higher hydrodynamic weight. Neither the existence of this exact difference nor the formal coefficient gives a uniform terminal estimate for it. Under (2.5) and the additional homothety the coefficient becomes $-12\nu^2 C^{-2}\tau^{-2-2h}$ and the remainder is multiplied by ν^4/ϵ^4, exactly as the curvature scaling proved above requires.

The actual zero-jet velocity gives a quantitative obstruction to a fixed ϵ endpoint use of the timelike hydrodynamic boundary frame. At the axis it is exactly $v_3 = \epsilon\sqrt{\nu}\, j_0\tau^{-1/2-h}$, so $v^2 < \nu$ requires

$$\epsilon^2 j_0^2 \tau^{-1-2h} < 1, \qquad \tau > (\epsilon j_0)^{1/(1/2+h)}. \tag{2.17}$$

The shrinking source swirl sample independently gives $v^2/\nu \geq \epsilon^2 e_0^2 \tau^{-1-2h}/4$ wherever the retained source remainder is at most $e_0/2$. The strain scale also grows as ϵ^2/τ. Thus the exact scalar coefficient (2.16) is a growing geometric image on its coefficient domain; it is not a proved spacetime singularity of an exact vacuum solution. The actual source force, its boundary representation, and its nonuniform terminal domain have all been computed rather than removed.

3 The actual field in the membrane force, shear, and entropy map

3.1 Source, conventions, and the complete parameter map

The source in this section is Eling–Fouxon–Oz, *The Incompressible Navier–Stokes Equations From Black Hole Membrane Dynamics*, arXiv:0905.3638v2. Its entire primary TeX was read, including the equations with source labels `bulkBB`, `shear`, `kappa`, `focusing`, `horizonNS`, and `areadiss`. These identify the imported geometrical equations precisely. The source derives a derivative expansion; it does not supply a convergent bulk construction for the present forced field. We calculate its coefficients, its matter projections, and its remainder identities. The source-existence conclusion for the actual fluid remains an import from the identified 166-page Navier–Stokes manuscript.

This section writes velocity arguments with time first. The coordinate permutation from the source's (x, t) notation is $(x, t) \mapsto (t, x)$, with inverse $(t, x) \mapsto (x, t)$; it does not alter any time value or spatial coordinate. Retain physical coordinates $(t, x) \in [0, 1) \times \mathbb{R}^3$, the source viscosity $\nu > 0$, and all original cutoffs, pressures, and forces. In particular

$$u_\nu(t, x) = \sqrt{\nu}\, u_*(t, x/\sqrt{\nu}), \qquad p_\nu(t, x) = \nu p_*(t, x/\sqrt{\nu}), \qquad f_\nu(t, x) = \sqrt{\nu}\, f_*(t, x/\sqrt{\nu}), \tag{3.1}$$

where u_* is the actual curl and azimuthal cutoff sum, rather than its leading profile. The equation is

$$\partial_t u_{\nu,i} + u_\nu^j \partial_j u_{\nu,i} + \partial_i p_\nu - \nu\Delta u_{\nu,i} = f_{\nu,i}, \qquad \partial_i u_\nu^i = 0. \tag{3.2}$$

The force includes every differentiation of those cutoffs.

EFO uses $8\pi G = c = \hbar = k_B = 1$. For three fluid dimensions the bulk dimension is five. Its black-brane scale and surface gravity are

$$T_0 = \frac{1}{4\pi\nu}, \qquad \kappa_0 = 2\pi T_0 = \frac{1}{2\nu}, \qquad a_H = \pi T_0, \qquad A_0 = a_H^3. \tag{3.3}$$

Here A_0 is an area density, not the infinite area of a planar horizon. The temperature map is invertible: $\nu = (4\pi T_0)^{-1}$. No viscosity is set to one in this map.

To make the source's order parameter explicit, for each $\varepsilon > 0$ define

$$\begin{aligned} &t = \varepsilon^2 s, && x = \varepsilon y, && v(s, y) = \varepsilon u_\nu(\varepsilon^2 s, \varepsilon y), \\ &P(s, y) = \varepsilon^2 p_\nu(\varepsilon^2 s, \varepsilon y), && F(s, y) = \varepsilon^3 f_\nu(\varepsilon^2 s, \varepsilon y). \end{aligned} \tag{3.4}$$

Its domain is $0 \le s < \varepsilon^{-2}$, $y \in \mathbb{R}^3$, and its inverse is

$$s = t/\varepsilon^2, \qquad y = x/\varepsilon,$$
$$u_\nu(t,x) = \varepsilon^{-1} v(t/\varepsilon^2, x/\varepsilon), \qquad p_\nu(t,x) = \varepsilon^{-2} P(t/\varepsilon^2, x/\varepsilon), \qquad f_\nu(t,x) = \varepsilon^{-3} F(t/\varepsilon^2, x/\varepsilon).$$

The time coordinate has been explicitly reparameterized, not identified with the original time. The three differentiated terms $\partial_s v$, $v \cdot \nabla_y v$, and $\nu \Delta_y v$, and the pressure gradient all acquire the same factor ε^3. Thus (v, P, F) obeys (3.2) with $t, x, u_\nu, p_\nu, f_\nu$ replaced by s, y, v, P, F, with the same ν. Its support is $\varepsilon^{-1} K_\nu$. This proves the coordinate and field maps, including their inverse.

All order statements below have the source's meaning $v, \partial_y = O(\varepsilon)$, $P, \partial_s = O(\varepsilon^2)$. They refer to coefficients evaluated on a fixed compact subset of $0 \le t < 1$ in the original variables. The constants may depend on that subset and on the full actual source field. They are not bounds uniform up to $t = 1$, and do not assert convergence of the metric series.

3.2 The geometric coefficient map and its remainders

Use EFO's horizon generator $n = \partial_s + v^i \partial_{y^i}$, whose normalization is $n^s = 1$, and its horizon first fundamental form

$$ds_H^2 = h_{ij}(dy^i - v^i ds)(dy^j - v^j ds), \qquad h_{ij} = a_H^2 \delta_{ij} + \rho_{ij}, \quad \rho_{ij} = O(\varepsilon^2). \tag{3.5}$$

Define the deformation tensor with the conventional factor one half, the expansion, and the trace-free shear by

$$B_{ij} = \tfrac{1}{2}(\partial_s h_{ij} + v^k \partial_k h_{ij} + h_{kj} \partial_i v^k + h_{ik} \partial_j v^k), \tag{3.6}$$
$$\theta = h^{ij} B_{ij}, \qquad \sigma_{ij} = B_{ij} - \tfrac{1}{3} \theta h_{ij}. \tag{3.7}$$

EFO calls the half-Lie-derivative tensor in (3.6) a Lie derivative in its first displayed transport formula; its subsequent shear formula has precisely the factor retained here. Write

$$S_{ij} = \tfrac{1}{2}(\partial_i v_j + \partial_j v_i), \qquad W_{ij} = \tfrac{1}{2}(\partial_i v_j - \partial_j v_i), \qquad v_j = \delta_{jk} v^k. \tag{3.8}$$

Spatial index contractions in $S : S$ use δ; contractions in $\sigma^2 = \sigma_{ij}\sigma_{k\ell} h^{ik} h^{j\ell}$ use h. These two metrics are retained separately.

Substitution of (3.5) into (3.6) gives the exact identity

$$\begin{aligned} B_{ij} &= a_H^2 S_{ij} + E_{ij}, \\ E_{ij} &= \tfrac{1}{2}(\partial_s \rho_{ij} + v^k \partial_k \rho_{ij} + \rho_{kj} \partial_i v^k + \rho_{ik} \partial_j v^k). \end{aligned} \tag{3.9}$$

Since $\partial_i v^i = 0$,

$$\theta = h^{ij}(a_H^2 S_{ij} + E_{ij}) = O(\varepsilon^4), \qquad \sigma_{ij} = a_H^2 S_{ij} + r_{ij}, \qquad r_{ij} = E_{ij} - \tfrac{1}{3}\theta h_{ij} = O(\varepsilon^4). \tag{3.10}$$

Indeed $E = O(\varepsilon^4)$, $h^{-1} = a_H^{-2} \delta^{-1} + O(\varepsilon^2)$, and the order-two trace is zero. In particular

$$\begin{aligned} \sigma^2 &= S : S + \mathcal{R}_\sigma, \\ \mathcal{R}_\sigma &= (h^{ik} h^{j\ell} - a_H^{-4} \delta^{ik} \delta^{j\ell}) a_H^4 S_{ij} S_{k\ell} \\ &\quad + h^{ik} h^{j\ell}(2 a_H^2 S_{ij} r_{k\ell} + r_{ij} r_{k\ell}) = O(\varepsilon^6). \end{aligned} \tag{3.11}$$

This is an exact remainder identity, rather than an equality between full shear and its leading coefficient.

The remaining EFO data are its surface gravity and normal connection:

$$\kappa = \kappa_0(1 + P - v^2/2) + r_\kappa, \qquad \Omega_i = \kappa_0 v_i + r_{\Omega,i}, \qquad r_\kappa = O(\varepsilon^4), \quad r_\Omega = O(\varepsilon^3). \tag{3.12}$$

They retain the exact velocity and pressure coefficients; corrections beyond the orders evaluated in the source are denoted by the displayed remainders. The extrinsic-curvature components in the basis (n, e_i) are $K_n^n = \kappa$, $K_i^n = \Omega_i$, and $K_j^i = \sigma_j^i + \theta \delta_j^i / 3$, as in EFO.

3.3 The force projection, including a discrepancy in the source

Let $\mathcal{T}_{AB}$ denote the right-hand side of EFO's bulk equation $R_{AB} + 4g_{AB} = \mathcal{T}_{AB}$. The curvature scale in this equation is one; the five-dimensional cosmological constant is $\Lambda = -6$, since the vacuum equation is $R_{AB} = 2\Lambda g_{AB}/3$. In natural units the relation to a physical matter tensor $T^{\rm phys}$ is the trace reversal

$$\mathcal{T}_{AB} = 8\pi G_5 \left(T^{\rm phys}_{AB} - \tfrac{1}{3} T^{\rm phys} g_{AB} \right).$$

With SI conventions the coupling is $8\pi G_5/c^4_{\rm light}$, and the trace and metric are both the physical ones in the same coordinate basis. EFO absorbs this coupling in its convention $8\pi G = 1$. The projections below are the coupling-weighted physical projections. Put

$$J_i = \mathcal{T}_{AB} n^A e_i^B, \qquad Q = \mathcal{T}_{AB} n^A n^B. \tag{3.13}$$

The cosmological term has zero contraction in both expressions because $g(n,n) = g(n,e_i) = 0$. These projections equal the physical matter projections times that coupling; adding a trace multiple of the metric to the matter tensor changes neither projection.

Retain EFO's displayed equation with source label `horizonNS`, including its printed signs:

$$(\partial_s + \theta)\Omega_i + v^j D_j \Omega_i + \Omega_j D_i v^j + \partial_i \kappa - D_j \sigma_i^j + \tfrac{1}{3}\partial_i \theta = -J_i. \tag{3.14}$$

Here D is the connection of h. Denote its left side by $\mathcal{C}_i$, and define, without deleting any term,

$$\mathcal{R}_i = \mathcal{C}_i - \left\{ \kappa_0 (\partial_s v_i + v^j \partial_j v_i + \partial_i P) - \tfrac{1}{2}\Delta v_i \right\}. \tag{3.15}$$

It follows that $\mathcal{R}_i = O(\varepsilon^5)$. To verify the coefficient, the two velocity-square derivatives cancel exactly:

$$\kappa_0 v_j \partial_i v^j - \tfrac{1}{2}\kappa_0 \partial_i (v_j v^j) = 0.$$

Also $\partial_j S_i^j = \frac{1}{2}\Delta v_i$, by incompressibility. The connection difference is order three and multiplies order-one Ω and v, while the remaining metric, expansion, and data corrections in (3.10)–(3.12) contribute order five. Equations (3.2) and (3.3) now give the exact residual identity

$$\boxed{J_i = -\kappa_0 F_i - \mathcal{R}_i, \qquad F_i = -\kappa_0^{-1}(J_i + \mathcal{R}_i).} \tag{3.16}$$

Thus the leading coefficient of the matter flux is $-2\pi T_0 F_i$. Its coordinate and force factors are those of (3.4); in original variables this coefficient is $-2\pi T_0 \varepsilon^3 f_{\nu,i}(t,x)$.

The equation immediately after `horizonNS` in EFO instead prints $f_i = \mathcal{T}_{AB} n^B e_i^A$. With the same $\Omega_i = 2\pi T_0 v_i$, the same pressure parametrization, and the same NS sign, that printed formula is incompatible with the displayed constraint by both a sign and the factor $2\pi T_0$. Equation (3.16) records the result of direct substitution into the displayed constraint. No silent change of stress units or orientation makes the two printed formulas identical.

The actual force is essential even modulo gradients. Indeed, suppose it were $f_\nu = \nabla\psi$ on the whole preterminal interval. Integration of (3.2) against u_ν, using its compact spatial support, gives

$$\frac{d}{dt}\frac{1}{2}\int |u_\nu|^2 dx + \nu \int |\nabla u_\nu|^2 dx = \int u_\nu \cdot \nabla\psi \, dx = 0.$$

The initial velocity is zero. Integration in time makes both nonnegative terms zero, hence $u_\nu = 0$. This contradicts the actual nonzero axis velocity computed below. Therefore the full forced construction cannot be made unforced by merely changing its pressure. In the coefficient map with these fixed flat spatial data, $\mathcal{T} = 0$ would require $F = 0$ at the retained order. This gives the precise obstruction and the explicit nonzero flux (3.16) that replaces it.

3.4 The independent null flux and its exact tensor fibres

The force projection does not determine the null flux Q. This is an explicit linear-algebra statement with an inverse, not an unproved absence of a relationship. At a horizon point choose the auxiliary null normal m with $g(n,m)=1$, orthogonal to the screen. Let ϑ^i be the screen covectors dual to e_i, annihilating n, m, and let $m^\flat = g(m,\cdot)$. For arbitrary scalar Q and screen covector $J = J_i\vartheta^i$, define

$$\mathcal{T}^{\mathrm{sec}} = Qm^\flat \otimes m^\flat + m^\flat \otimes J + J \otimes m^\flat. \tag{3.17}$$

Evaluation on (n,n) gives Q, and evaluation on (n,e_i) gives J_i, since $m^\flat(n)=1$, $m^\flat(e_i)=J(n)=0$. The complete fibre is

$$\mathcal{T} = \mathcal{T}^{\mathrm{sec}} + \mathcal{K}, \qquad \mathcal{K}(n,n) = \mathcal{K}(n,e_i) = 0. \tag{3.18}$$

Conversely subtracting (3.17) from every tensor with those projections gives exactly this kernel. This proves the section, inverse recovery of (Q,J), and the full fibre. It is a pointwise tensor construction; neither conservation nor a bulk solution of a specified matter model is implied.

There is also an exact energy-condition obstruction. For a screen vector X and real z, the vector

$$k(z) = n + zX - \tfrac{1}{2}z^2 h(X,X)m$$

is null: its squared norm is $z^2h(X,X) - z^2h(X,X) = 0$. Consequently

$$\mathcal{T}(k(z),k(z)) = Q + 2zJ(X) + O(z^2).$$

If $Q=0$ and the null energy condition holds for both signs of small z, then $J(X)=0$ for every X. Thus a nonzero force projection cannot coexist with $Q=0$ under that condition. The converse pointwise realizability for $Q>0$ can be written explicitly. Define the covector ℓ by

$$\ell(n) = 1, \qquad \ell(e_i) = J_i/Q, \qquad \ell(m) = -\frac{h^{ij}J_iJ_j}{2Q^2}.$$

Its metric norm is zero, and $\mathcal{T} = Q\ell\otimes\ell$ has the requested projections and is nonnegative on every null vector. Retain also every complementary component of this particular section. Writing $|J|_h^2 = h^{ij}J_iJ_j$, they are

$$\begin{aligned} &\mathcal{T}(e_i,e_j) = J_iJ_j/Q, && \mathcal{T}(n,m) = -|J|_h^2/(2Q),\\ &\mathcal{T}(m,e_i) = -|J|_h^2 J_i/(2Q^2), && \mathcal{T}(m,m) = |J|_h^4/(4Q^3). \end{aligned}$$

These follow by multiplying the corresponding two evaluations of ℓ and the factor Q. When J and Q have nonzero leading orders three and four, respectively, these components have orders two, two, one, and zero. Thus even this explicit null-energy-positive section need not be a small perturbation of vacuum in all its components. This is a property of the displayed section, not an obstruction proved for every possible matter model. The algebraic maps do not replace the unsupplied bulk matter dynamics.

3.5 The actual axis strain and the normal-connection curvature

The full axis-gradient proof is (1.12). Its essential source data are also specified here to fix the actual input: source Lemma 5.1, page 47, prescribes both $U_n(0,\eta) = \varphi_n(0,\eta) = 0$ for every positive order; source (B.1), page 144, fixes $U_{\mathrm{axis}}(\eta) = 4\eta + j_0$, with $0 < j_0 \le 0.05$; source (B.3), page 145, gives $\varphi_*(0) = 1$. The positive-order sums are locally finite before time one, the annular corrections vanish near the axis, and the final spatial and temporal cutoffs equal one near the origin for all sufficiently small $\tau = 1 - t > 0$. For the source's unchanged $C > 1$, $h > 0$, and $A = \frac{1}{2} + h$, the resulting exact physical identities are

$$u_\nu(1-\tau, 0) = (0, 0, \sqrt{\nu}\, j_0 \tau^{-A}),$$
$$G := D_x u_\nu(1-\tau, 0) = \begin{pmatrix} -2/\tau & -b(\tau) & 0 \\ b(\tau) & -2/\tau & 0 \\ 0 & 0 & 4/\tau \end{pmatrix}, \qquad b(\tau) = C^{-1}\tau^{-1-h}. \tag{3.19}$$

The viscosity independence of G is the cancellation of the two factors $\sqrt{\nu}$ and $1/\sqrt{\nu}$ in (3.1), not a viscosity-one replacement. In particular, the symmetric and antisymmetric contractions are

$$\left|\tfrac{1}{2}(G + G^{\mathsf{T}})\right|^2 = \frac{24}{\tau^2}, \qquad |\operatorname{curl} u_\nu|^2 = \frac{4}{C^2\tau^{2+2h}}. \tag{3.20}$$

The first identity sums $4+4+16$; the second uses $\partial_1 u_2 - \partial_2 u_1 = 2b$. Thus the exact growing rotation cancels from the symmetric tensor, while the independently calculated diagonal entries produce the nonzero strain. The squared full velocity derivative is $24\tau^{-2} + 2C^{-2}\tau^{-2-2h}$.

Evaluate the horizon coefficients at $s_\tau = (1-\tau)/\varepsilon^2$, $y = 0$. From (3.4), every spatial velocity derivative gains ε^2. Therefore the shear image is

$$\sigma^2(s_\tau, 0) = \frac{24\varepsilon^4}{\tau^2} + \mathcal{R}_\sigma(s_\tau, 0). \tag{3.21}$$

This is the source's coefficient calculation with the exact remainder (3.11) still present.

The rotation has its own exact geometric map. Define the curvature two-form of the screen normal connection by

$$\mathcal{F}_{ij} = \partial_i \Omega_j - \partial_j \Omega_i = \kappa_0(\partial_i v_j - \partial_j v_i) + \partial_i r_{\Omega,j} - \partial_j r_{\Omega,i}. \tag{3.22}$$

To verify that this is normal-connection curvature, the normal projection satisfies $\nabla_i^\perp n = \Omega_i n$ and $\nabla_i^\perp m = -\Omega_i m$, by metric compatibility and the two null norms. Computing the commutator on n gives $[\nabla_i^\perp, \nabla_j^\perp] n = \mathcal{F}_{ij} n$. Under a change of null normalization $n' = e^\lambda n$, $m' = e^{-\lambda} m$, direct differentiation gives $\Omega'_i = \Omega_i + \partial_i \lambda$, and hence $\mathcal{F}' = \mathcal{F}$. The contraction $\mathcal{F}_{ij}\mathcal{F}_{k\ell}h^{ik}h^{j\ell}$ is therefore invariant under both screen coordinate changes and this normal-frame gauge. The chosen horizon slicing is retained.

At the actual axis its coefficient is

$$\begin{aligned} \mathcal{F}_{12}(s_\tau, 0) &= \frac{2\kappa_0 \varepsilon^2}{C\tau^{1+h}} + (dr_\Omega)_{12}(s_\tau, 0), \\ |\mathcal{F}|_h^2(s_\tau, 0) &= \frac{8\kappa_0^2 \varepsilon^4}{a_H^4 C^2 \tau^{2+2h}} + \mathcal{R}_{\mathcal{F}}(s_\tau, 0). \end{aligned} \tag{3.23}$$

The factor eight is twice the square of the displayed coefficient of $\mathcal{F}_{12}$; the two nonzero entries are 12 and 21. Precisely, if $F_{ij}^0 = \kappa_0(\partial_i v_j - \partial_j v_i)$ and $E^F = dr_\Omega$, the remainder is

$$\begin{aligned} \mathcal{R}_{\mathcal{F}} = {} & (h^{ik}h^{j\ell} - a_H^{-4}\delta^{ik}\delta^{j\ell}) F_{ij}^0 F_{k\ell}^0 \\ & + h^{ik}h^{j\ell}(2F_{ij}^0 E_{k\ell}^F + E_{ij}^F E_{k\ell}^F). \end{aligned}$$

It is order six on each of the stated compact preterminal sets. Equation (3.23) computes a geometrical normal-bundle curvature coefficient. It does not identify this contraction with a bulk Riemann curvature scalar.

3.6 Forced area and entropy, with the measure and error retained

The imported EFO focusing equation in bulk dimension five is

$$-n(\theta) + \kappa\theta - \theta^2/3 - \sigma^2 = Q. \tag{3.24}$$

With the shear remainder already proved, rearrange it exactly as

$$\begin{aligned} \kappa_0\theta &= S : S + Q + \mathcal{R}_{\mathrm{foc}}, \\ \mathcal{R}_{\mathrm{foc}} &= \mathcal{R}_\sigma + n(\theta) + \theta^2/3 - (\kappa - \kappa_0)\theta. \end{aligned} \tag{3.25}$$

The incompressible order hierarchy has $Q = O(\varepsilon^4)$; an order-two Q would instead alter its order-two focusing equation. In this hierarchy $\mathcal{R}_{\mathrm{foc}} = O(\varepsilon^6)$, because n differentiates at order two, $\theta = O(\varepsilon^4)$, and $\kappa - \kappa_0 = O(\varepsilon^2)$. In particular the actual shear contribution to the expansion at the axis is

$$\theta(s_\tau, 0) = \frac{24\varepsilon^4}{\kappa_0\tau^2} + \frac{Q(s_\tau, 0)}{\kappa_0} + \frac{\mathcal{R}_{\mathrm{foc}}(s_\tau, 0)}{\kappa_0} = \frac{48\nu\varepsilon^4}{\tau^2} + \frac{Q + \mathcal{R}_{\mathrm{foc}}}{\kappa_0}(s_\tau, 0). \tag{3.26}$$

The total expansion is not determined by the velocity alone: the complete missing fibre is (3.18), and its relation to the force and to the null energy condition was proved above.

For an exact finite-area statement choose a bounded smooth spatial domain D_ε containing the fixed support $\varepsilon^{-1}K_\nu$ in its interior. Let $d\mu_h = \sqrt{\det h}\, d^3y$ and $A_D(s) = \int_{D_\varepsilon} d\mu_h$. The determinant derivative identity and (3.6) give

$$\theta\sqrt{\det h} = \partial_s\sqrt{\det h} + \partial_i(v^i\sqrt{\det h}).$$

Consequently the general boundary formula is

$$\frac{dA_D}{ds} = \int_{D_\varepsilon} \theta\, d\mu_h - \int_{\partial D_\varepsilon} \sqrt{\det h}\, v^i N_i\, dS_{\mathrm{Eucl}}. \tag{3.27}$$

The boundary integral is exactly zero for the actual compactly supported velocity. No statement about an infinite total planar area is needed. The same derivative applies to the finite excess $A_D - A_0|D_\varepsilon|$. Substituting (3.25) yields

$$\frac{dA_D}{ds} = \frac{A_0}{\kappa_0}\int_{D_\varepsilon} (S : S + Q)\, d^3y + \mathcal{R}_A, \tag{3.28}$$

$$\mathcal{R}_A = \frac{1}{\kappa_0}\int_{D_\varepsilon} \Big[(\sqrt{\det h} - A_0)(S : S + Q) + \sqrt{\det h}\,\mathcal{R}_{\mathrm{foc}}\Big]\, d^3y. \tag{3.29}$$

The integrand remainder is order six on the specified compact sets. The domain grows as ε^{-1}, so an integrated order-six bound without its volume factor is not asserted.

For completeness the exact strain integral is

$$\begin{aligned} \int_{D_\varepsilon} S : S\, d^3y &= \frac{1}{2}\int_{D_\varepsilon} \partial_i v_j \partial_i v_j\, d^3y + \frac{1}{2}\int_{D_\varepsilon} \partial_i v_j \partial_j v_i\, d^3y \\ &= \frac{1}{2}\int_{D_\varepsilon} |\nabla v|^2\, d^3y. \end{aligned}$$

For the last equality integrate the cross term by parts in i. The boundary term vanishes because v is zero near the boundary, and the remaining integral is $-\int v_j\partial_j(\partial_i v_i) = 0$.

Put $E(s) = \frac{1}{2}\int_{D_\varepsilon} |v|^2 d^3y$. Multiplying the complete forced NS equation by v and integrating gives

$$\frac{dE}{ds} + \nu\int_{D_\varepsilon} |\nabla v|^2 d^3y = \int_{D_\varepsilon} v \cdot F\, d^3y. \tag{3.30}$$

The convective integral is the integral of $\operatorname{div}(v|v|^2/2)$; the pressure integral is $-\int P \operatorname{div} v$; the Laplacian integral is $-\int |\nabla v|^2$. This proves every term and sign in (3.30). Combining it with $1/(2\kappa_0) = \nu$ gives

$$\boxed{\frac{1}{A_0}\frac{dA_D}{ds} = -\frac{dE}{ds} + \int_{D_\varepsilon} v \cdot F\, d^3y + \frac{1}{\kappa_0}\int_{D_\varepsilon} Q\, d^3y + \frac{\mathcal{R}_A}{A_0}.} \tag{3.31}$$

The work term is indispensable for the actual source. Under (3.4), $E(s) = \varepsilon^{-1}E_\nu(t)$, $dE/ds = \varepsilon\, dE_\nu/dt$, and $\int v \cdot F\, d^3y = \varepsilon \int u_\nu \cdot f_\nu\, d^3x$, since $d^3y = \varepsilon^{-3}d^3x$. These factors recover the original energy and force exactly.

The Bekenstein–Hawking entropy for this portion of the horizon is $\mathscr{S}_D = A_D/(4G)$, equal to $2\pi A_D$ in the source's units, where this G is the five-dimensional Newton constant in that convention. Its leading unperturbed entropy density is $s_0 = A_0/(4G)$. The local entropy production density along the generators is exactly $\sqrt{\det h}\,\theta/(4G)$. Its fourth-order coefficient is

$$\frac{s_0}{\kappa_0}(S:S+Q) = \frac{A_0}{T_0}(S:S+Q), \qquad (8\pi G = 1). \tag{3.32}$$

The remainder is explicitly $\big[(\sqrt{\det h} - A_0)(S:S+Q) + \sqrt{\det h}\mathcal{R}_{\rm foc}\big]/(4G\kappa_0)$. At the actual axis the shear contribution to (3.32) is

$$\frac{24 s_0 \varepsilon^4}{\kappa_0 \tau^2} = \frac{48\nu s_0 \varepsilon^4}{\tau^2}. \tag{3.33}$$

The full density retains the additional $s_0 Q/\kappa_0$ and the displayed measure and focusing remainders. Equation (3.31), divided by $4G$, is its integrated entropy and force-work relation. In the unforced, zero-null-flux coefficient it reduces to EFO's area–energy formula with its exact area density and viscosity; for the actual forced field the two additional physical terms are retained.

Here is an explicit inverse restoration of the entropy units. Fix the gravitational length unit $\ell_g > 0$, write $g^{\rm phys} = \ell_g^2 g$, and retain physical constants $c_{\rm light}, \hbar, k_B, G_5$. A three-dimensional horizon cross-section then has $A_D^{\rm phys} = \ell_g^3 A_D$, and its physical entropy is

$$S_D^{\rm phys} = \frac{k_B c_{\rm light}^3}{4\hbar G_5} A_D^{\rm phys} = \frac{k_B c_{\rm light}^3 \ell_g^3}{8\pi\hbar G_5}\mathscr{S}_D, \qquad \mathscr{S}_D = \frac{8\pi\hbar G_5}{k_B c_{\rm light}^3 \ell_g^3} S_D^{\rm phys}. \tag{3.34}$$

The first area equality follows because multiplication of a three-dimensional induced metric by ℓ_g^2 multiplies its volume form by ℓ_g^3. Thus all entropy rates above are restored by an explicit positive constant, as well as the selected time-coordinate conversion for a rate per physical time. For the normalization $t_g^{\rm phys} = (\ell_g/c_{\rm light})s$, the temperature assigned to this generator is

$$T_H^{\rm phys} = \frac{\hbar c_{\rm light}}{k_B \ell_g} T_0, \qquad T_0 = \frac{k_B \ell_g}{\hbar c_{\rm light}} T_H^{\rm phys}.$$

An observer's proper-time temperature also carries its specified Tolman redshift. Neither T_0 nor the original numerical ν is identified with an SI quantity without this unit and time dictionary.

3.7 The separate Rindler background and its temperature

The preceding black-brane background has cosmological constant $\Lambda = -6$ at unit curvature scale. EFO's Rindler example has $\Lambda = 0$ and a different screen metric. Its exact unperturbed line element, with a separate static time coordinate τ_R, is

$$ds_R^2 = -\kappa_0^2 \xi^2 d\tau_R^2 + d\xi^2 + \delta_{ij}dy^i dy^j, \qquad \xi > 0. \tag{3.35}$$

It is flat by the explicit map $X^0 = \xi \sinh(\kappa_0 \tau_R)$, $X^4 = \xi \cosh(\kappa_0 \tau_R)$, $X^i = y^i$. The inverse on $X^4 > |X^0|$ is $\xi = ((X^4)^2 - (X^0)^2)^{1/2}$, $\tau_R = \kappa_0^{-1} \operatorname{arctanh}(X^0/X^4)$. Direct differentiation gives $-(dX^0)^2 + (dX^4)^2 = -\kappa_0^2 \xi^2 d\tau_R^2 + d\xi^2$. Eddington–Finkelstein coordinates are

$$\begin{aligned} r_R &= \kappa_0 \xi^2, & s &= \tau_R + \kappa_0^{-1} \log(\xi/\xi_{\text{ref}}), \\ \xi &= (r_R/\kappa_0)^{1/2}, & \tau_R &= s - \kappa_0^{-1} \log\left((r_R/\kappa_0)^{1/2}/\xi_{\text{ref}}\right), \end{aligned} \tag{3.36}$$

where the positive reference length ξ_{ref} is retained and changes only the additive time origin. Substitution gives $ds_R^2 = -\kappa_0 r_R ds^2 + ds\, dr_R + dy^2$. The screen at $r_R = 0$ has $h_{ij}^{R,(0)} = \delta_{ij}$, area density $A_0^R = 1$, and $\kappa_0 = 2\pi T_0$. These are Rindler data, not the replacement of πT_0 by one in a black-brane solution.

For an observer at fixed ξ, its lapse is $\chi = \kappa_0 \xi$, so proper time is $d\tau_{\text{prop}} = \chi d\tau_R$. The normalized tangent is $U = (\kappa_0 \xi)^{-1} \partial_{\tau_R}$. The connection coefficient $\Gamma^{\xi}_{\tau_R \tau_R} = \kappa_0^2 \xi$ gives $\nabla_U U = \xi^{-1} \partial_\xi$, hence proper acceleration $a = 1/\xi$. The source-imported Unruh temperature law gives

$$T_{\text{loc}} = \frac{a}{2\pi} = \frac{1}{2\pi\xi} = \frac{T_0}{\chi}, \qquad \xi = \frac{1}{2\pi T_{\text{loc}}}, \qquad T_0 = \chi T_{\text{loc}}. \tag{3.37}$$

The reference observer $\xi = \kappa_0^{-1}$ has $\chi = 1$, so precisely for that observer $T_0 = a_0/(2\pi)$, with $a_0 = \kappa_0$. Restoring units gives $a^{\text{phys}} = c_{\text{light}}^2/(\ell_g \xi)$ and $T_{\text{loc}}^{\text{phys}} = \hbar a^{\text{phys}}/(2\pi k_B c_{\text{light}})$, with the inverses given by these positive factors.

EFO applies its null deformation and focusing analysis separately to slow Rindler perturbations, obtaining the same numerical viscosity $(4\pi T_0)^{-1}$. On this branch the intrinsic proofs (3.6)–(3.11) and (3.24)–(3.31) use $h^R = \delta + \rho^R$, their own remainders, and $A_0^R = 1$. They therefore give the same actual shear coefficient $24\varepsilon^4/\tau^2$, the same expansion coefficient $48\nu\varepsilon^4/\tau^2$, and entropy density with $s_0^R = 1/(4G_5)$. This follows by repeating the displayed contractions with the Rindler screen metric, whose inverse is δ, and not by equating the two bulk metrics. The source's Rindler example imposes $R_{AB} = 0$, so its matter projections are zero. Applying the actual forced field through matter projections changes that vacuum hypothesis. A force represented instead by the boundary geometry is the separate explicit map of Section 2; it is not a deduction of nonzero matter from the flat Rindler vacuum itself.

3.8 The actual range of the expansion

The field maps (3.1) and (3.4), the fluid identities, and the actual axis data hold at every preterminal point in their stated domains. The derivative expansion has a smaller range. In fact (3.19) gives the exact necessary slow-motion constraint, for any fixed admissible velocity threshold $\delta > 0$,

$$|v(s_\tau, 0)| = \varepsilon\sqrt{\nu}\, j_0 \tau^{-A} \le \delta \quad \Longrightarrow \quad \tau \ge \left(\frac{\varepsilon\sqrt{\nu}\, j_0}{\delta}\right)^{1/A}. \tag{3.38}$$

Moreover the exact rotation derivative has magnitude $\varepsilon^2/(C\tau^{1+h})$, and its ratio to κ_0 is $2\nu\varepsilon^2/(C\tau^{1+h})$. These explicitly calculated quantities already leave the source's small-velocity and small-derivative regime at fixed $\varepsilon > 0$ as $\tau \downarrow 0$. Additional higher derivatives must still obey the orders retained by the chosen expansion; (3.38) alone is not a sufficient convergence criterion.

Accordingly (3.21), (3.23), and (3.33) establish the exact images of the actual fluid coefficients and their explicit preterminal growth. They retain $\mathcal{R}_\sigma$, $\mathcal{R}_{\mathcal{F}}$, $\mathcal{R}_{\text{foc}}$, the horizon measure, the force work, and Q. Taking $\tau \downarrow 0$ at fixed ε in only their leading terms cannot establish a bulk endpoint singularity or an exact thermodynamic endpoint. The computed maps and the quantitative obstruction (3.38) specify exactly what the source correspondence does establish for this construction.

4 The actual forced field on a round sphere and the Schwarzschild boundary constraint

4.1 An invertible flux map with the physical coordinates retained

Throughout this section $u = u_\nu$, $p = p_\nu$, and $f = f_\nu$ are the actual cutoff-summed, globally localized source fields, with the original physical viscosity $\nu > 0$ and time $0 \le t < 1$. Their imported defining formulas are source (9.21), (10.4)–(10.5), (10.11), and (10.22) [1, pp. 115, 118, 120, 124]. In particular u, p, f are smooth before time one, u, p have one fixed compact spatial support, $u(0) = 0$, and

$$\partial_t u^i + u^j \partial_j u^i + \partial_i p - \nu \Delta u^i = f^i, \qquad \partial_i u^i = 0. \tag{4.1}$$

The source's proof of existence and terminal growth is imported, rather than claimed as an independent verification here. Every transformation below is proved directly from these exact fields.

Fix $L > 0$. Write $y = x$ for the unchanged physical Cartesian coordinate, and set

$$D = L^2 + |y|^2, \qquad \Omega = \frac{2L^2}{D}, \qquad \Phi_L(y) = \left(\frac{2L^2 y}{D}, \frac{L(|y|^2 - L^2)}{D}\right). \tag{4.2}$$

This maps $\mathbb{R}^3$ onto $S^3_L \setminus \{N\} \subset \mathbb{R}^4$, where $N = (0,0,0,L)$. Indeed its squared Euclidean length is $[4L^4|y|^2 + L^2(|y|^2 - L^2)^2]/D^2 = L^2$. For $X = (X^1, X^2, X^3, X^4) \in S^3_L \setminus \{N\}$ the inverse is

$$y^i = \frac{LX^i}{L - X^4}. \tag{4.3}$$

Substitution into both compositions gives the identity. Differentiating (4.2) and taking ambient scalar products gives $\partial_i \Phi_L \cdot \partial_j \Phi_L = \Omega^2 \delta_{ij}$. Thus the round metric and its volume form in these coordinates are exactly

$$g_{ij} = \Omega^2 \delta_{ij}, \qquad dV_g = \Omega^3 d^3 y, \qquad \mathrm{Vol}(S^3_L) = 2\pi^2 L^3. \tag{4.4}$$

The volume follows by integrating the displayed density: with $r = L \tan a$, $4\pi \int_0^\infty 8L^6 r^2 (L^2 + r^2)^{-3} dr = 32\pi L^3 \int_0^{\pi/2} \sin^2 a \cos^2 a$
$, da = 2\pi^2 L^3$.

Define the new field and the retained pressure by

$$a = \Omega^{-3}, \qquad v^i(t, \Phi_L(y)) = a(y) u^i(t, y), \qquad P_0(t, \Phi_L(y)) = p(t, y). \tag{4.5}$$

Here v^i are coefficients in the coordinate basis $\partial_i \Phi_L$; the ambient tangent vector is $(D\Phi_L)_y(au)$. The identity $\iota_v dV_g = \iota_u d^3 y$ proves preservation of flux, not preservation of Euclidean speed. In coordinates it gives the complete divergence proof

$$\nabla_i v^i = \Omega^{-3} \partial_i (\Omega^3 v^i) = \Omega^{-3} \partial_i u^i = 0. \tag{4.6}$$

Outside the fixed compact support of u, p, both fields in (4.5) vanish. Their extensions by zero near N are therefore smooth on the entire sphere for every $t < 1$. Conversely any smooth pair (v, P_0) vanishing on a fixed neighbourhood of N has the exact inverse $u^i(y) = \Omega^3(y) v^i(\Phi_L(y))$ and $p(y) = P_0(\Phi_L(y))$; (4.6) proves that divergence zero is equivalent in both directions. There is no quotient of the pressure or change of the physical viscosity.

4.2 All conformal derivative terms and the exact force

Use Euclidean contractions in the following coordinate expressions, and put

$$q_i = \partial_i \log \Omega = -\frac{2y_i}{D}, \quad \partial_j q_i = -\frac{2\delta_{ij}}{D} + \frac{4y_i y_j}{D^2}, \quad \Delta \log \Omega = -\frac{6}{D} + \frac{4|y|^2}{D^2}. \tag{4.7}$$

The metric formula for the Levi–Civita connection gives

$$\Gamma^i_{jk} = \delta^i_j q_k + \delta^i_k q_j - \delta_{jk} q_i. \tag{4.8}$$

Inserting it in the coordinate definition of Ricci gives $R_{ij} = -\partial_i q_j + q_i q_j - (\Delta \log \Omega + |q|^2)\delta_{ij}$ in dimension three. Using (4.7), its $y_i y_j$ coefficients cancel and its remaining coefficient is $8L^2/D^2 = 2\Omega^2/L^2$. Hence

$$R_{ij} = \frac{2}{L^2} g_{ij}, \qquad R^i{}_j = \frac{2}{L^2}\delta^i_j. \tag{4.9}$$

For an arbitrary vector z, expansion of two covariant derivatives yields

$$(\nabla^2 z)^i = \Omega^{-2}\big[\Delta z^i + 3q^j \partial_j z^i + 2q_j \partial_i z^j - 2q^i \partial_j z^j + (\Delta \log \Omega + |q|^2) z^i - q^i (q_j z^j)\big]. \tag{4.10}$$

For clarity, this follows from $g^{jk}(\partial_j \nabla_k z^i + \Gamma^i_{j\ell} \nabla_k z^\ell - \Gamma^\ell_{jk} \nabla_\ell z^i)$: the three terms contribute respectively $\Delta z^i + q_j \partial_i z^j + q^j \partial_j z^i + (\Delta \log \Omega) z^i - q^i \partial_j z^j$, $q_j \partial_i z^j + q^j \partial_j z^i - q^i \partial_j z^j - q^i (q_j z^j)$, and $q^j \partial_j z^i + |q|^2 z^i$, inside the common factor Ω^{-2}. This proves (4.10), including all its signs.

Since $\partial_i a = -3a q_i$ and $\partial_i u^i = 0$, define

$$\begin{aligned}\mathcal{V}^i[u] &= (\nabla^2 v)^i + R^i{}_j v^j \\ &= \Omega^{-5}\big[\Delta u^i - 3q^j \partial_j u^i + 2q_j \partial_i u^j + (-2\Delta \log \Omega + |q|^2 + 2\Omega^2/L^2) u^i - q^i (q_j u^j)\big].\end{aligned} \tag{4.11}$$

To check the substitution explicitly, $\Delta(au) = a[\Delta u - 6q \cdot \partial u + (-3\Delta \log \Omega + 9|q|^2)u]$ and $\partial_i(a u^i) = -3a(q \cdot u)$. These identities and (4.10) give (4.11) term by term. Likewise (4.8) gives

$$v^j \nabla_j v^i = a^2\{u^j \partial_j u^i - (q_j u^j) u^i - q^i |u|^2\}, \qquad \nabla^i P_0 = \Omega^{-2} \partial_i p. \tag{4.12}$$

The complete force on the sphere is the following explicit field:

$$\begin{aligned}G^i &= a f^i + (a^2 - a) u^j \partial_j u^i + (\Omega^{-2} - a)\partial_i p + \nu a \Delta u^i - \nu \mathcal{V}^i[u] \\ &\quad - a^2\{(q_j u^j) u^i + q^i |u|^2\}.\end{aligned} \tag{4.13}$$

It is smooth for $t < 1$ and zero in a fixed neighbourhood of N. Substitute (4.1) in $\partial_t v = a \partial_t u$ and then use (4.11)–(4.12). Every term in (4.13) is obtained without a remainder, proving

$$\partial_t v^i + v^j \nabla_j v^i + \nabla^i P_0 - \nu(\nabla^2 v^i + R^i{}_j v^j) = G^i, \qquad \nabla_i v^i = 0. \tag{4.14}$$

The differential operator is exactly the one in [4, §2, eq. (1)]; here its right side has been calculated for the actual source field. Its inverse force map is explicit: write the right side of (4.13) as $af + \mathcal{Q}[u, p]$, where $\mathcal{Q}$ is the entire displayed expression after the first term. Then

$$f^i(y) = \Omega^3(y)\big(G^i(\Phi_L(y)) - \mathcal{Q}^i[\Omega^3 v, P_0](y)\big). \tag{4.15}$$

Together with (4.3) and (4.5), this is an inverse on triples satisfying the corresponding equations. In particular the geometric terms are retained data and have not been treated as zero. The smooth terminal extension of the original f does not by itself give a smooth terminal extension of G: (4.13) also contains the growing velocity and its derivatives.

4.3 Invariant concentration, dissipation, and a proved pressure obstruction

At every chart point the exact metric speed and energy are

$$|v|_g^2 = \Omega^{-4}|u|^2, \qquad E_g(t) = \frac{1}{2}\int_{S_L^3} |v|_g^2 dV_g = \frac{1}{2}\int_{\mathbb{R}^3} \Omega^{-1}|u|^2 d^3y. \tag{4.16}$$

Since the support is fixed, Ω^{-1} is bounded there, so the source's bounded Euclidean energy implies bounded E_g before time one. At the actual source sampling point $y_\tau = (\sqrt{2\nu X_{\rm in}\tau}, 0, 0)$, the unit tangent in its second coordinate direction is $e_{(2)} = \Omega^{-1}\partial_2\Phi_L$. The signed, geometrically defined component is exactly

$$g(v, e_{(2)})(1-\tau, \Phi_L(y_\tau)) = \left(\frac{L^2 + 2\nu X_{\rm in}\tau}{2L^2}\right)^2 \sqrt{\nu}\,\tau^{-1/2-h}(e_0 + R_*(\tau)), \tag{4.17}$$

where the original remainder satisfies $R_*(\tau) = O(\tau^{2h})$, by source (10.20)–(10.23) [1, p. 124]. For sufficiently small positive τ, its absolute value is at least $(\sqrt{\nu}e_0/8)\tau^{-1/2-h}$, because $\Omega^{-2} \geq 1/4$ and $|e_0 + R_*| \geq e_0/2$. This proves unbounded invariant speed on the sphere, not merely unbounded coordinate coefficients.

Define $S_{ij}(v) = (\nabla_i v_j + \nabla_j v_i)/2$ and use the usual two-form norm $|F|_g^2 = \frac{1}{2}g^{ik}g^{j\ell}F_{ij}F_{k\ell}$. From $v_i = \Omega^{-1}u_i$ and (4.8) one gets

$$S_{ij}(v) = \Omega^{-1}\left[\frac{\partial_i u_j + \partial_j u_i}{2} - \frac{3}{2}(q_i u_j + q_j u_i) + \delta_{ij}(q\cdot u)\right], \tag{4.18}$$

$$(dv^\flat)_{ij} = \Omega^{-1}[\partial_i u_j - \partial_j u_i - q_i u_j + q_j u_i]. \tag{4.19}$$

For example the covariant derivative before taking its symmetric part is $\nabla_i v_j = \Omega^{-1}[\partial_i u_j - 2q_i u_j - q_j u_i + \delta_{ij}(q\cdot u)]$, proving both formulas. Contracting (4.18) gives zero trace as required by (4.6). At $y = 0$, $\Omega = 2$ and $q = 0$. Inserting the actual source axis jet (1.12), namely

$$(\partial_j u^i)(1-\tau, 0) = \begin{pmatrix} -2/\tau & -b & 0 \\ b & -2/\tau & 0 \\ 0 & 0 & 4/\tau \end{pmatrix}, \qquad b = C^{-1}\tau^{-1-h}, \tag{4.20}$$

proves the two scalar identities

$$|S(v)|_g^2(1-\tau, \Phi_L(0)) = \frac{3}{8\tau^2}, \qquad |dv^\flat|_g^2(1-\tau, \Phi_L(0)) = \frac{1}{16C^2}\tau^{-2-2h}. \tag{4.21}$$

Indeed the squared Euclidean norm of the symmetric matrix in (4.20) is $24/\tau^2$; the two-form with entries $F_{12} = 2b$, $F_{21} = -2b$ has squared two-form norm $4b^2$. The conformal contraction in either case multiplies by $\Omega^{-6} = 1/64$. Thus the original constant C and the source exponent are retained. The same axis input gives the additional exact speed formula

$$|v|_g^2(1-\tau, \Phi_L(0)) = \frac{\nu j_0^2}{16}\tau^{-1-2h}, \tag{4.22}$$

because $u(1-\tau, 0) = \sqrt{\nu}j_0\tau^{-1/2-h}e_3$ and $\Omega^{-4}(0) = 1/16$. All axis identities hold on the source's fixed interval $0 < \tau < \tau_1$, where its localization is identically one, as established in (1.12). For the spatial metric itself the exact curvature image is also known. Its unit normal in the explicit embedding (4.2) is Φ_L/L, whose derivative is $L^{-1}D\Phi_L$. The Euclidean Gauss equation, obtained by decomposing the ambient derivative into its tangent and normal parts and commuting its two derivatives, therefore gives $R_{ijkl} = L^{-2}(g_{ik}g_{j\ell} - g_{i\ell}g_{jk})$. In an orthonormal basis its contraction gives

$$R_g = 6/L^2, \qquad |\,\mathrm{Ric}_g\,|_g^2 = 12/L^4, \qquad |\,\mathrm{Riem}_g\,|_g^2 = 12/L^4. \tag{4.23}$$

For the last contraction, the three-dimensional Kronecker expression has squared norm $2\cdot 3\cdot 2 = 12$. These identities calculate the curvature of the actual spatial target metric at the concentrating points as well as everywhere else. The growing scalar derivatives in (4.21) are properties of its velocity field.

There is also an exact global obstruction to erasing the force. For any divergence-free v on this sphere,

$$\nabla_j(2S^{ij}) = \nabla^2 v^i + R^i{}_j v^j. \tag{4.24}$$

To verify it, expand $2S^{ij} = \nabla^i v^j + \nabla^j v^i$ and commute $\nabla_j \nabla^i v^j = \nabla^i \nabla_j v^j + R^i{}_j v^j$; the first term is zero. The commutator follows by substituting (4.8) into the definition of curvature. Multiply (4.14) by v_i, integrate, and apply the divergence theorem on the closed sphere. Convection integrates to $\int \nabla_j(v^j|v|^2/2) = 0$, pressure to $\int \nabla_j(P_0 v^j) = 0$, and (4.24) gives

$$\frac{d}{dt}E_g(t) + 2\nu \int_{S^3_L} |S(v)|^2_g dV_g = \int_{S^3_L} g(G, v) dV_g. \tag{4.25}$$

Here the antisymmetric part of ∇v contracts to zero against S. The original zero initial velocity gives $E_g(0) = 0$. Therefore for every $t < 1$ at which the actual source field is nonzero,

$$\int_0^t \int_{S^3_L} g(G, v) dV_g, dt' = E_g(t) + 2\nu \int_0^t \int_{S^3_L} |S(v)|^2_g dV_g, dt' > 0. \tag{4.26}$$

Such times exist by (4.17). Hence G cannot vanish identically. More strongly it cannot equal $\nabla\zeta$ for a smooth global scalar $\zeta(t,\cdot)$ for all times: that would make the left side zero by incompressibility. No scalar pressure change can turn this actual time-dependent field into an unforced sphere solution on the whole interval. This proves the exact obstruction relevant to the vacuum ansatz below; it makes no statement excluding other gravitational boundary data or bulk matter.

The nonnegative local viscous conversion rate appearing in the kinetic energy balance is $2\nu|S(v)|^2_g$, and at the source axis it is exactly $3\nu/(4\tau^2)$. If a positive prescribed temperature T_b and constant mass density ρ_b are supplied as thermodynamic data, the specified viscous entropy-production observable $2\rho_b\nu|S(v)|^2_g/T_b$ is exactly $3\rho_b\nu/(4T_b\tau^2)$ there. This last expression defines that observable with its units and constitutive inputs; it does not derive a temperature field or horizon entropy law from incompressibility.

4.4 The source time, the literal metric, and its finite-order domain

The viscosity-one variables used by Bredberg–Strominger [4, §2] are related to the retained variables by

$$s = \nu t, \qquad w^i(s) = \nu^{-1} v^i(s/\nu), \qquad \Pi_0(s) = \nu^{-2} P_0(s/\nu), \qquad \mathfrak{G}^i(s) = \nu^{-2} G^i(s/\nu). \tag{4.27}$$

All spatial coordinates and L are unchanged. Its inverse is $t = s/\nu$, $v(t) = \nu w(\nu t)$, $P_0(t) = \nu^2 \Pi_0(\nu t)$, $G(t) = \nu^2 \mathfrak{G}(\nu t)$. Every term in (4.14) acquires the common factor ν^2, proving

$$\partial_s w^i + w^j \nabla_j w^i + \nabla^i \Pi_0 - (\nabla^2 w^i + R^i{}_j w^j) = \mathfrak{G}^i. \tag{4.28}$$

The interval is $0 \le s < \nu$, so terminal time one has not been identified with source time one. This is a reversible map, not deletion of ν.

For a three-dimensional fluid the source spacetime has dimension five. Write its mass parameter as $m > 0$, retain its near-horizon parameter $\alpha > 0$, and set $L = 2m$ only for the leading round spatial metric in this construction. Its Schwarzschild exterior has $F(r) = 1 - (2m/r)^2$ and metric $-F(r)d\tau_{\rm Schw}^2 + F(r)^{-1}dr^2 + r^2 d\Omega_3^2$. The source coordinate change [4, §3, eq. (3)] and its explicit exterior inverse are

$$\begin{aligned} s &= \frac{\alpha}{m}\left[\tau_{\rm Schw} + r_*(r)\right], & \rho &= \frac{m}{\alpha}(r-2m), \\ r &= 2m + \frac{\alpha}{m}\rho, & \tau_{\rm Schw} &= \frac{m}{\alpha}s - r_*(r), \\ r_*(r) &= r + m\log\frac{r-2m}{r+2m} + c_*, & r &> 2m. \end{aligned} \tag{4.29}$$

Differentiating r_* gives $1/F$. Substituting $d\tau_{\rm Schw} = (m/\alpha)ds - dr/F$ and $dr = (\alpha/m)d\rho$ cancels the dr^2/F term and gives $-(m/\alpha)^2F$
$, ds^2 + 2ds$
$, d\rho + r^2 d\Omega_3^2$ exactly. The coordinates extend smoothly to the future horizon $\rho = 0$; the inverse involving $\tau_{\rm Schw}$ has only its stated exterior domain. The boundary is $\rho = 1$, at $r_c = 2m + \alpha/m$, and is not the leading radius $2m$.

With $A_\alpha = 1 + \alpha/(2m^2)$, the source parameter is exactly

$$\lambda = \frac{1}{4K^2} = \frac{m^2A_\alpha^2(1 - A_\alpha^{-2})}{(3 - 2A_\alpha^{-2})^2} = \alpha + O(\alpha^2). \tag{4.30}$$

This is the $p = 3$ specialization of source (5), not an exact substitution $\lambda = \alpha$. Since the displayed rational function is analytic near zero with derivative one, it has a unique local analytic inverse there. Equivalently that inverse is specified without a discarded term by the root $A_\alpha > 1$ near one of $\lambda(3A_\alpha^2 - 2)^2 = m^2A_\alpha^4(A_\alpha^2 - 1)$, followed by $\alpha = 2m^2(A_\alpha - 1)$.

Here is the source metric (6) with dimension fixed to three and its fields replaced by the actual mapped fields. Set $\Pi = \Pi_0 + c(s)$, keep $g = 4m^2d\Omega_3^2$ as the leading round metric for contractions, and distinguish the source tensor Λ_{ij} from the scalar λ:

$$ds^2_{\rm bulk} = -\frac{\rho}{\lambda}ds^2 + \frac{6\rho(5+\rho)}{8m^2}ds^2 + 2ds \tag{4.31}$$

$$, d\rho + 4m^2d\Omega_3^2$$

$$+ (1-\rho)(|w|_g^2ds^2 - 2w_ids \tag{4.32}$$

$$, dy^i) - 2\rho\Pi ds^2$$

$$\begin{aligned} &+ \lambda\big[(4\rho + 8m^2\Pi)d\Omega_3^2 + (1-\rho)w_iw_jdy^idy^j \\ &\qquad - (\rho^2 - 1)(\nabla^2 w_i + 2R_{ji}w^j)ds \end{aligned} \tag{4.33}$$

$$, dy^i - 2w_id\rho \tag{4.34}$$

$$, dy^i + (|w|_g^2 + 2\Pi)ds \tag{4.35}$$

$$, d\rho$$

$$+ 2(1-\rho)\phi_ids \tag{4.36}$$

$$, dy^i\big] + \lambda^2(2\chi_idy^id\rho + \Lambda_{ij}dy^idy^j) + \mathcal{R}_{\rm metric}. \tag{4.37}$$

The term $2R_{ji}w^j$ inside this metric is literally the source's $(p-1)R_{ji}w^j$. Its coefficient is not changed to the coefficient one in the Navier–Stokes operator (4.28). $\mathcal{R}_{\rm metric}$ denotes the components not supplied by source (6); the source explicitly allows further corrections at order λ or above which do not change its Einstein-equation calculation through order λ^0. It therefore does not assert a uniform $O(\lambda^3)$ bound for this entire tensor, and no such bound is assigned here. Likewise the round radius-r_c reference boundary metric differs from the leading radius $L = 2m$ by its retained $O(\lambda)$ correction.

There is an internal sign discrepancy in the printed source which is relevant to its timelike boundary reference. At $\rho = 1$ its metric (6), also retained in (4.37), has $g_{ss} = -1/\lambda + 9/(2m^2) -$

$2\Pi + O(\lambda)$, whereas its product metric (8) prints $-[1/\lambda + 9/(2m^2) + O(\lambda)]ds^2$ before the conformal factor. These constant terms have opposite signs. The exact Schwarzschild coordinate calculation above fixes the sign for the background used here: expansion of (4.30) gives $\lambda = \alpha - 15\alpha^2/(4m^2) + O(\alpha^3)$, while at $\rho = 1$ $-(m/\alpha)^2[1-(1+\alpha/(2m^2))^{-2}] = -1/\alpha + 3/(4m^2) + O(\alpha)$. Their substitution gives $-1/\lambda + 9/(2m^2) + O(\lambda)$, agreeing with source (6). Thus source (8) is reported with its discrepancy, and is not used as an exact equality to the written metric. Its spatial conformal factor, used below, has no such sign change.

Source §4 requires incompressibility at order λ^{-1} and the *unforced* viscosity-one equation at order λ^0. Our incompressibility constraint is solved exactly by (4.6); our momentum expression at the next order is the explicit nonzero $\mathfrak{G}$ in (4.28). Thus (4.37) gives a typed substitution in the source ansatz and identifies its failed vacuum momentum constraint. By (4.26), its failure cannot be removed by changing the scalar pressure. This computation has not constructed a vacuum Einstein solution with those data, nor determined the bulk matter or changed boundary metric that would solve a different Einstein problem. The exact force (4.13) is the datum such a further construction would have to represent. The source establishes its unforced construction only through Einstein-equation order λ^0 and explicitly does not prove all-order existence for the sphere [4, §4]. In particular neither fixed-λ convergence up to $t = 1$ nor a spacetime curvature singularity follows from (4.21).

4.5 The complete global pressure mode and an explicit divergence inverse

Even at the displayed finite order the source imposes an integral condition on its pressure [4, §4, eqs. (9), (11)]. We calculate this condition without overwriting the original p. Let $V_L = 2\pi^2 L^3$ and fix any reference $s_0 \in [0, \nu)$ and scalar c_0. Define the auxiliary geometric pressure mode by

$$c(s) = c_0 - \frac{1}{3V_L}\left[\int_{S^3_L} (|w(s)|^2_g - |w(s_0)|^2_g)dV_g \right.$$
$$\left. + 3\int_{S^3_L} (\Pi_0(s) - \Pi_0(s_0))dV_g\right]. \tag{4.38}$$

It is smooth for $s < \nu$ and has value c_0 at s_0. Since $\Pi = \Pi_0 + c$, differentiation proves exactly

$$3\partial_s \int_{S^3_L} \Pi\, dV_g + \partial_s \int_{S^3_L} |w|^2_g dV_g = 0. \tag{4.39}$$

Conversely this identity and $c(s_0) = c_0$ integrate to (4.38), so this auxiliary mode is unique. A constant factor L^3 converts $d\Omega_3$ in the source's (11) to dV_g; it cancels from the equation. The inverse back to the physical source pressure remains explicit: $p(t, y) = \nu^2[\Pi(\nu t, \Phi_L(y)) - c(\nu t)]$.

There is a direct global solution to the displayed scalar-divergence equation itself. Put

$$\begin{aligned} H &= 3\partial_s\Pi + \partial_s|w|^2_g, \\ J^i &= 2\Pi w^i - \tfrac{1}{2}\nabla^2 w^i - w^j\nabla_j w^i - w^j\nabla^i w_j. \end{aligned} \tag{4.40}$$

Then $\int H\, dV_g = 0$ by (4.39). For points X, Y on S^3_L, let $\chi = d_g(X, Y)/L$ and use the integrable kernel

$$\mathcal{K}_L(\chi) = -\frac{1}{4\pi^2 L}\left[(\pi - \chi)\cot\chi - \frac{1}{2}\right]. \tag{4.41}$$

It satisfies the distributional identity $\Delta_X \mathcal{K}_L(d_g(X, Y)/L) = \delta_Y - V_L^{-1}$. Here is a complete check. For a radial function the metric in geodesic polar coordinates is $L^2(d\chi^2 + \sin^2\chi\, d\Omega_2^2)$; the divergence formula gives $\Delta F = L^{-2}(F'' + 2\cot\chi F')$. Away from zero, direct differentiation of (4.41) gives $\Delta\mathcal{K}_L = -1/V_L$. At zero, $\mathcal{K}_L = -1/(4\pi L\chi) + O(1)$, so its outward radial derivative has flux tending to one through the small geodesic two-sphere. At $\chi = \pi$ the product $(\pi - \chi)\cot\chi$ is a smooth even function of $\pi - \chi$ with value -1, so there is no other singularity. These facts, integration by parts on a punctured sphere, and the limiting unit flux prove the stated distributional identity. Finally $\int_0^\pi (\pi - \chi)\sin\chi\cos\chi\,d\chi = \pi/4$ proves that the subtracted $1/2$ makes $\int \mathcal{K}_L\,dV_g = 0$.

Define

$$\Psi(s,X) = \int_{S^3_L} \mathcal{K}_L(d_g(X,Y)/L)H(s,Y)dV_g(Y), \qquad \phi^i = J^i + \nabla^i\Psi. \tag{4.42}$$

The kernel is locally integrable and H is smooth. To see smoothness without differentiating its singularity, fix a point X_0 and choose smooth local rotations $R_X \in SO(4)$ with $R_X X_0 = X$. Change variables $Y = R_X Z$ in the integral; the kernel becomes the fixed $\mathcal{K}_L(d_g(X_0,Z)/L)$ and all derivatives fall on $H(s,R_X Z)$. Compactness and integrability justify differentiating to every order. The distributional identity and $\int H = 0$ now give $\Delta\Psi = H$ pointwise. Consequently

$$\nabla_i\phi^i = 3\partial_s\Pi + \partial_s|w|_g^2 + \nabla_i J^i, \tag{4.43}$$

the exact right side printed in source (9) for $p = 3$. This explicitly solves its global mean obstruction as a scalar equation. Source (9) was derived after using the unforced equation; for our actual forced input this scalar identity alone does not imply the remaining Einstein equations or change the momentum obstruction already proved.

The source boundary conformal factor is $1 + 2\lambda\Pi + O(\lambda^2)$ and its reference spatial radius is $r_c = 2m + O(\lambda)$ [4, §4, eqs. (7)–(8)]. On every compact interval $s \le s_* < \nu$, the associated formal spatial volume coefficient is therefore

$$\mathrm{Area}(\Sigma_s) = 2\pi^2 r_c^3 + 3\lambda \int_{S^3_{r_c}} \Pi(s)\,dV_{r_c} + O_{s_*}(\lambda^2). \tag{4.44}$$

Indeed multiplying a three-metric by $1 + 2\lambda\Pi$ multiplies its volume form by $(1+2\lambda\Pi)^{3/2} = 1 + 3\lambda\Pi + O_{s_*}(\lambda^2)$. If the unknown source remainder has no actual uniform realization, this equation is interpreted solely as its coefficient identity. The fixed r_c is independent of s, and $dV_{r_c} = (r_c/L)^3 dV_g$, so (4.39) gives

$$[\partial_s\,\mathrm{Area}(\Sigma_s)]_\lambda = -\left(\frac{r_c}{L}\right)^3 \partial_s \int_{S^3_L} |w|_g^2 dV_g = \left(\frac{r_c}{L}\right)^3 \left[4\int |S(w)|_g^2 dV_g - 2\int g(\mathfrak{G},w)dV_g\right]. \tag{4.45}$$

The bracket on the left denotes the coefficient multiplying λ; the final equality follows by rescaling (4.25) exactly. At a strictly Taylor-expanded coefficient one sets $r_c/L = 1$; retaining the displayed ratio instead keeps the finite boundary-radius relation and changes only terms of order λ^2 in (4.44). This is the boundary-area coefficient of the written ansatz with its chosen pressure mode, not the area of a proved vacuum event horizon. The source itself describes the possible entropy-area interpretation as a suggestion [4, §4]. Thus the exact computed work term, axis dissipation, and boundary-volume coefficient remain explicit, while no uncontrolled terminal or horizon identification is made.

5 Actual thermal observables, Hamiltonian transport, and modular time

This section uses the actual localized, cutoff-summed field of the 166-page released manuscript [1], rather than replacing it by a leading profile. Its existence and all-order residual construction are source-imported conclusions; the maps and thermodynamic calculations below are proved here. The additional thermometric model is specified explicitly. In particular its temperature is an observable assigned by that model, and is not an equation of state supplied by the Navier–Stokes manuscript.

5.1 Source fields and the exact axis observable

Write the source's actual locally summed potentials and pressure as $A_*, B_* e_\theta, p_{*,\mathrm{loc}}$, including its finite initialization and all terms of (9.21), p. 115. With the exact source cutoff $c_* = \chi_x \chi_t$, equations (10.4)–(10.5), p. 118, are

$$u_* = \operatorname{curl}(c_* A_*) + c_* B_* e_\theta, \qquad p_* = c_* p_{*,\mathrm{loc}}, \qquad f_* = \partial_t u_* + (u_* \cdot \nabla) u_* - \Delta u_* + \nabla p_*.$$

Every derivative of every cutoff belongs to this force. Its extension through $t = 1$ is the source's (10.11). For each original $\nu > 0$, use exactly (10.22), p. 124:

$$y = x/\sqrt{\nu}, \quad u_\nu(x,t) = \sqrt{\nu}\, u_*(y,t), \quad p_\nu(x,t) = \nu p_*(y,t), \quad f_\nu(x,t) = \sqrt{\nu}\, f_*(y,t).$$

The inverse is $x = \sqrt{\nu} y$, with velocity and force multiplied by $\nu^{-1/2}$, and pressure by ν^{-1}. Time is unchanged. Thus

$$\partial_t u_\nu + u_\nu \cdot \nabla u_\nu - \nu \Delta u_\nu + \nabla p_\nu = f_\nu, \qquad \nabla \cdot u_\nu = 0. \tag{5.1}$$

This follows term by term from the chain rule: the four terms on the left are $\sqrt{\nu}$ times their starred counterparts. The field has fixed compact spatial support $K_\nu = \sqrt{\nu} K_*$, and is zero on the entire initial time interval $0 \le t \le 1 - \tau_0$.

Here is the axis calculation from the source data. Retain $0 < h < 1/100$, the positive source amplitude C, and $\tau = 1 - t$. Its similarity coordinates satisfy

$$\tau = q(1 - \eta^2), \qquad z = q^{1/2-h}\eta, \qquad X = r^2/(2q).$$

At $z = 0$ the inverse gives $\eta = 0, q = \tau$. Source (5.1), p. 46, and Lemma 5.1, p. 47, give

$$\frac{u_{\theta,n}}{r} = C^{-1} q^{-1-h+2nh} \varphi_n(X,\eta), \qquad \varphi_n(0,\eta) = 0 \quad (n \ge 1).$$

Proposition 5.5, pp. 60–61, realizes these coefficients with cutoffs $\chi(c_n q)$, retaining order zero. Since $q > 0$, $c_n \to \infty$, and χ has compact support, the sum is locally finite at every preterminal point. Its axis value therefore is exactly $C^{-1} q^{-1-h} \varphi_0(0,\eta)$. Source (B.3) and (B.12), pp. 145–147, give

$$\varphi_0(0,\eta) = \varphi_*(\eta), \qquad \varphi_*(\eta) = \exp\left(\Lambda \int_0^\eta \zeta_*(w)\, dw\right), \qquad \varphi_*(0) = 1.$$

The function ζ_* here is the source's auxiliary axis function. All annular representatives in (9.21) vanish near the axis at each $t < 1$; the base potential is azimuthal and its curl supplies only radial and axial velocity. Finally the localization cutoffs equal one near $(x,t) = (0,1)$. Consequently a source-determined $\tau_1 > 0$ exists such that the *actual* field satisfies

$$\left.\frac{u_{\theta,*}(r,0,1-\tau)}{r}\right|_{r=0} = C^{-1}\tau^{-1-h} \qquad (0 < \tau < \tau_1).$$

This is evaluation of the smooth quotient, not division by zero. For $r_* = r/\sqrt{\nu}$, the physical quotient is $\sqrt{\nu} u_{\theta,*}(r_*, z_*, t)/(\sqrt{\nu} r_*)$. Thus it has the same axis value.

In the axis neighborhood the actual field is smooth and axisymmetric, so it can be written

$$u_1 = a(r^2,z,t)x_1 - b(r^2,z,t)x_2, \qquad u_2 = b(r^2,z,t)x_1 + a(r^2,z,t)x_2, \qquad u_3 = d(r^2,z,t).$$

Direct differentiation at $x = 0$ gives $\partial_1 u_2 = b$, $\partial_2 u_1 = -b$, and zero horizontal components of the curl. Hence the original Euclidean vorticity obeys

$$\omega_\nu(0,1-\tau) := \operatorname{curl} u_\nu(0,1-\tau) = \left(0, 0, \frac{2}{C}\tau^{-1-h}\right), \qquad |\omega_\nu(0,1-\tau)|^2 = \frac{4}{C^2}\tau^{-2-2h}. \tag{5.2}$$

The norm is unchanged by an orientation-preserving orthonormal change of the original spatial frame. This is a fluid observable; it is not being identified with a spacetime curvature invariant. Every positive-order axis value vanished exactly, so no remainder has been dropped. At the distinct source sample $x_\tau = (\sqrt{2\nu X_{\mathrm{in}}\tau}, 0, 0)$, the actual formula remains

$$(u_\nu)_2(x_\tau, 1-\tau) = \sqrt{\nu}\tau^{-1/2-h}(e_0 + R_*(\tau)), \qquad |R_*(\tau)| \le C_*\tau^{2h}.$$

No derivative estimate for this last remainder is inferred from its zeroth-order bound.

5.2 The retained mechanical-energy and heat balances

Retain laboratory length and time units $\ell_f, t_f > 0$ and a constant laboratory mass density $\rho_f > 0$. The complete unit map is

$$x_{\mathrm{lab}} = \ell_f x, \qquad t_{\mathrm{lab}} = t_f t, \qquad u_{\mathrm{lab}} = \frac{\ell_f}{t_f} u_\nu,$$
$$p_{\mathrm{lab}} = \frac{\ell_f^2}{t_f^2} p_\nu, \quad \nu_{\mathrm{lab}} = \frac{\ell_f^2}{t_f}\nu, \quad f_{\mathrm{lab}} = \frac{\ell_f}{t_f^2} f_\nu.$$

Here p_{lab} is pressure per unit mass; multiplication by ρ_f gives physical pressure. The inverse divides each coordinate or field by its displayed nonzero factor. Write $\varrho = \rho_f \ell_f^5/t_f^2$, the energy coefficient for integration in the original dx measure, and set

$$S_{ij} = \tfrac{1}{2}(\partial_j u_{\nu,i} + \partial_i u_{\nu,j}), \qquad e = \tfrac{\varrho}{2}|u_\nu|^2, \quad \mathcal{J}_j = (e + \varrho p_\nu)u_{\nu,j} - 2\varrho\nu u_{\nu,i}S_{ij}.$$

Repeated spatial indices in this subsection are summed from one to three. Incompressibility implies $2\partial_j S_{ij} = \Delta u_{\nu,i}$. Multiplying (5.1) by $\varrho u_{\nu,i}$, applying the product rule to convection, pressure, and stress, gives the exact pointwise identity

$$\partial_t e + \partial_j \mathcal{J}_j = \varrho f_\nu \cdot u_\nu - 2\varrho\nu S_{ij}S_{ij}. \tag{5.3}$$

For the last contraction, write $\partial_j u_{\nu,i} = S_{ij} + W_{ij}$, with $W_{ij} = -W_{ji}$; then $S_{ij}W_{ij} = 0$ by pairing the two indices. This proves the coefficient and sign of the last term.

One precise isothermal model couples this mechanical field to a bath held at a specified $T_{\mathrm{b}} > 0$, withdrawing the local viscous power $q_{\mathrm{b}} = 2\varrho\nu|S|^2$ per source time and coordinate volume. The bath entropy delivered by this power, with derivative taken in the original source time t, is, by this model's definition,

$$\dot{\mathscr{S}}_{\mathrm{b}}(t) = \frac{2\varrho\nu}{T_{\mathrm{b}}}\int_{\mathbb{R}^3} |S(x,t)|^2\, dx \ge 0.$$

This specifies the thermostat and its measured entropy rather than supplying an unproved temperature equation for the original fluid. Compact support removes all boundary fluxes at spatial infinity. Integration by parts also gives

$$\int \partial_j u_{\nu,i}\partial_i u_{\nu,j}\, dx = -\int u_{\nu,i}\partial_i(\partial_j u_{\nu,j})\, dx = 0, \qquad 2\int |S|^2\, dx = \int |\nabla u_\nu|^2\, dx.$$

Write $F_\nu(t) = \int_0^t \|f_\nu(s)\|_2\, ds$. The integrated equation is

$$\tfrac{1}{2}\partial_t\|u_\nu\|_2^2 + \nu\|\nabla u_\nu\|_2^2 = \langle f_\nu, u_\nu\rangle.$$

Dividing by $(\|u_\nu\|_2^2 + \epsilon^2)^{1/2}$, integrating, and sending $\epsilon \downarrow 0$, proves $\|u_\nu(t)\|_2 \le F_\nu(t)$. Integrating the equation once more and using this bound proves

$$\|u_\nu(t)\|_2^2 + 2\nu\int_0^t \|\nabla u_\nu(s)\|_2^2 ds \le F_\nu(t)^2.$$

Here $F_\nu(1) = \nu^{5/4}F_*(1) < \infty$, directly from $dx = \nu^{3/2}dy$ and $f_\nu = \sqrt{\nu}f_*$. Monotone convergence in the nonnegative dissipation integral therefore proves

$$0 \le \mathscr{Q}_{\mathrm{b}} := 2\varrho\nu \int_0^1 \int |S|^2 \, dx \, dt \le \frac{\varrho}{2}\nu^{5/2}F_*(1)^2, \qquad \Delta\mathscr{S}_{\mathrm{b}} = \mathscr{Q}_{\mathrm{b}}/T_{\mathrm{b}}. \tag{5.4}$$

This is also the physical heat:

$$\mathscr{Q}_{\mathrm{b}} = 2\rho_f\nu_{\mathrm{lab}} \int_0^{t_f} \int_{\mathbb{R}^3} |S_{\mathrm{lab}}|^2 \, dx_{\mathrm{lab}} \, dt_{\mathrm{lab}}.$$

Indeed $S_{\mathrm{lab}} = S/t_f$, and the complete product of conversion factors is $\rho_f(\ell_f^2/t_f)\ell_f^3 t_f/t_f^2 = \rho_f\ell_f^5/t_f^2 = \varrho$. The antisymmetric axis rotation in (5.2) does not itself give S, and hence does not by itself give a pointwise lower bound on viscous heating. Equations (5.2) and (5.4) calculate the two specified observables in their actual pointwise and integrated domains.

5.3 The full material flow has an exact Hamiltonian lift

For any closed interval strictly below one the source field is smooth, has compact support, and has bounded spatial derivative. The integral equation for $\dot{x} = u_\nu(x,t)$ is a contraction on sufficiently short subintervals; its unique solutions concatenate, and bounded velocity prevents escape on the closed interval. Denote the resulting two-time map by $X_{t,s}(x_s)$. Uniqueness proves

$$X_{t,s}X_{s,r} = X_{t,r}, \qquad X_{t,s}^{-1} = X_{s,t}.$$

Differentiating the integral equation gives $\partial_t DX_{t,s} = Du_\nu(X_{t,s},t)DX_{t,s}$. Jacobi's determinant identity and incompressibility give $\det DX_{t,s} = 1$, beginning from the identity at $t = s$.

On $T^*\mathbb{R}^3$ with its original canonical coordinates (x,p), use the time-dependent Hamiltonian

$$H_{\mathrm{lift}}(t,x,p) = p_i u_{\nu,i}(x,t).$$

Hamilton's equations are exactly

$$\dot{x}_i = u_{\nu,i}(x,t), \qquad \dot{p}_i = -p_j\partial_i u_{\nu,j}(x,t).$$

The solution and its inverse are

$$(x_s,p_s) \longmapsto (X_{t,s}(x_s), DX_{t,s}(x_s)^{-\mathsf{T}}p_s), \qquad (x_t,p_t) \longmapsto (X_{s,t}(x_t), DX_{t,s}(X_{s,t}x_t)^{\mathsf{T}}p_t). \tag{5.5}$$

Indeed differentiating the matrix inverse proves the momentum equation and direct substitution proves both inverse compositions. Also $p_t \cdot dx_t = p_s \cdot dx_s$, so this map preserves the canonical one-form and its exterior derivative.

Retain the complete force and pressure through the material acceleration of the x projection:

$$\ddot{x}_i = \nu\Delta u_{\nu,i}(X_{t,s}x_s,t) - \partial_i p_\nu(X_{t,s}x_s,t) + f_{\nu,i}(X_{t,s}x_s,t).$$

Thus this lift carries the full given forced motion; it has not changed its equation to unforced motion. It describes material particles and their cotangent variables, not an independently derived microscopic Hamiltonian for the fluid.

There is also an exact autonomous lift. On $T^*((0,1)\times\mathbb{R}^3)$, put

$$K(t,x,p_0,p) = p_0 + p\cdot u_\nu(x,t).$$

Its evolution parameter λ satisfies $dt/d\lambda = 1$. The spatial and covector maps are (5.5), with $t = s + \lambda$, and

$$p_0(t) = p_0(s) + p_s\cdot u_\nu(x_s,s) - p_t\cdot u_\nu(x_t,t).$$

Differentiation gives $\dot p_0 = -p \cdot \partial_t u_\nu$; the terms involving Du_ν cancel using the other Hamilton equations. This proves the full lifted solution. Its inverse uses $\lambda \mapsto -\lambda$ and the same displayed conservation identity. Its domain is precisely the preterminal times for which both endpoints belong to $(0,1)$; a global extension across the source terminal time is not asserted.

This exact autonomous lift has a concrete Gibbs obstruction. For every positive inverse energy parameter $\beta_{\text{phys}} > 0$, its proposed canonical partition integral with Liouville measure contains

$$\int_{\mathbb{R}} \exp\{-\beta_{\text{phys}}[p_0 + p \cdot u_\nu(x,t)]\}\, dp_0 = \infty.$$

The integrand is positive, so Tonelli's theorem proves divergence of the full partition integral on any positive-measure set of (t,x,p). Thus the unrestricted canonical lift does not supply a normalizable Gibbs state.

Finally no fixed algebra identification and constant nonzero time rescaling can turn the original material evolution into a one-parameter autonomous automorphism group for its whole initial interval. If that were possible, its pullback on spatial observables would be the identity for an open interval of group parameters, because $u_\nu = 0$ near time zero. A group which is the identity on an open interval containing zero is the identity everywhere: for any parameter a, choose an integer n with a/n in that interval and take the n-fold product. It would follow that $X_{t,0}$ is always the identity (smooth coordinate test functions separate points), hence $u_\nu = 0$, contradicting (5.2). The two-time flow and the autonomous cotangent lift above are the exact correspondences which remain available.

5.4 A represented Bost–Connes system with all time conventions

The source for the next algebra is Bost–Connes [6], Sections 5–6, especially Proposition 24, equations (3)–(5), and Theorem 25 in Section 6 (IHES preprint printed pp. 32–33; published pp. 441–442). Fix the identity embedding of the roots of unity. Its representation acts on $\mathcal{H} = \ell^2(\mathbb{N}^*)$, with orthonormal basis e_m, by

$$\mu_n e_m = e_{nm}, \qquad v(r)e_m = e^{2\pi i m r} e_m \quad (n, m \in \mathbb{N}^*,\ r \in \mathbb{Q}/\mathbb{Z}).$$

Write $\mathcal{A} = C^*(\mu_n, v(r)) \subset B(\mathcal{H})$. The map from the source abstract algebra to $\mathcal{A}$ is its stated representation π_1; no unproved faithfulness or inverse to that representation is needed here. All subsequent algebra identifications are on this fully specified represented algebra. For example

$$\mu_n^* e_m = \begin{cases} e_{m/n}, & n \mid m, \\ 0, & n \nmid m, \end{cases} \quad v(r)\mu_n = \mu_n v(nr), \quad \mu_n v(r) \mu_n^* = \frac{1}{n} \sum_{ns=r} v(s).$$

For the last equality, both sides vanish on e_m when $n \nmid m$, by the finite sum of n-th roots of unity; when $m = nq$ both sides multiply e_m by $e^{2\pi i q r}$. These calculations also retain the denominators in the source relations.

Let $Le_m = (\log m)e_m$, on

$$D(L) = \left\{ a : \sum_{m \geq 1} (\log m)^2 |a_m|^2 < \infty \right\}.$$

This diagonal real operator is self-adjoint: the adjoint domain is exactly the stated weighted square-summability domain, by testing each e_m. Its source dynamics and inverse are

$$\sigma_v(A) = e^{ivL} A e^{-ivL}, \qquad \sigma_v^{-1} = \sigma_{-v}, \qquad \sigma_v(\mu_n) = n^{iv}\mu_n, \quad \sigma_v(v(r)) = v(r). \tag{5.6}$$

These identities follow by applying both sides to every e_m. They prove that σ_v preserves $\mathcal{A}$. The group is norm-continuous on each finite word in the generators, and hence on $\mathcal{A}$, by norm approximation and isometry.

There is a local sign inconsistency in the printed source: the sentence preceding Section 5 equation (2) writes $\sigma_t(e_X) = k^{-it}e_X$ with $k = R(X)/L(X)$, whereas that equation gives $\sigma_t(\mu_n) = n^{it}\mu_n$. The latter agrees with Proposition 4 and with the explicit Section 6 covariance. Equation (5.6) fixes the convention by its direct operator calculation, while retaining the source discrepancy.

The weak closure $\mathcal{A}''$ equals $B(\mathcal{H})$. To prove this, an operator commuting with every $v(r)$ has zero (m, n) entry when $m \neq n$: choose rational r for which $e^{2\pi imr} \neq e^{2\pi inr}$. It is therefore diagonal. Commutation with μ_n, applied to e_1, makes every diagonal entry equal to its first entry. Thus $\mathcal{A}'$ consists exactly of scalars and taking commutants proves the assertion.

For the dimensionless parameter $b > 1$, define

$$Z(b) = \sum_{m\geq 1} m^{-b} = \zeta(b), \qquad \rho_b = Z(b)^{-1}e^{-bL}, \qquad \phi_b(A) = \mathrm{Tr}(\rho_b A). \tag{5.7}$$

The integral comparison with $\int_1^\infty x^{-b}dx$ proves that $Z(b) < \infty$ precisely for $b > 1$. The eigenvalues $p_m = m^{-b}/Z(b)$ are positive and sum to one. Thus ϕ_b is a faithful normal state on $B(\mathcal{H})$: if $A \geq 0$ and $\phi_b(A) = 0$, every $\|A^{1/2}e_m\|^2$ is zero, and then $A = 0$.

For completeness its KMS property can be checked without importing a classification theorem. For finite matrices put $F_{A,B}(z) = \phi_b(B\sigma_z(A))$. This is entire, is bounded on $0 \leq \mathrm{Im}\, z \leq b$, and cyclicity of the finite trace gives

$$F_{A,B}(v + ib) = \phi_b(\sigma_v(A)B).$$

Equivalently, on matrix units $E_{mn} = |e_m\rangle\langle e_n|$, $\sigma_{ib}(E_{mn}) = (m/n)^{-b}E_{mn}$, so

$$\phi_b(E_{nm}\sigma_{ib}(E_{mn})) = p_n(m/n)^{-b} = p_m = \phi_b(E_{mn}E_{nm}).$$

Here is the extension from finite matrices to all bounded A, B. Let P_M project onto the first M basis vectors, and replace A, B by P_MAP_M, P_MBP_M. Strong convergence of these compressions and their adjoints, their uniform operator bound, and the finite-rank approximation of the trace-class ρ_b imply convergence in both seminorms $\|A\rho_b^{1/2}\|_2$ and $\|A^*\rho_b^{1/2}\|_2$. These seminorms are invariant under σ_v. Cauchy–Schwarz for the trace therefore proves uniform convergence of both boundary functions in real v. The difference of two compressed functions is bounded and holomorphic on the strip; the three-lines maximum estimate bounds its interior supremum by its two boundary suprema. They are consequently uniformly Cauchy on the entire closed strip. Their limit is continuous on its boundary, holomorphic inside, and has the displayed KMS boundary values. Restricting to $\mathcal{A}$ proves the claimed b-KMS state for σ.

The associated modular construction is completely explicit. On the Hilbert space $\mathcal{K}$ of Hilbert–Schmidt operators, represent $B(\mathcal{H})$ by left multiplication and use $\Omega_b = \rho_b^{1/2}$. The vector is cyclic because $E_{mn}\Omega_b = \sqrt{p_n}E_{mn}$ spans all finite matrices, which are dense in $\mathcal{K}$. It is separating because $A\Omega_b = 0$ implies $Ae_n = 0$ for every n. The Tomita operator on its initial core has

$$\begin{aligned} S_bE_{mn} &= \sqrt{p_m/p_n}\, E_{nm} && \text{antilinearly,} \\ JE_{mn} &= E_{nm} && \text{antilinearly,} \\ \Delta_bE_{mn} &= (p_m/p_n)E_{mn}. \end{aligned}$$

The diagonal operator Δ_b has domain $\sum_{m,n}(p_m/p_n)^2|a_{mn}|^2 < \infty$; its positive square root has the corresponding first-power weight. Finite matrices form a core by truncating the weighted sum. This also identifies the closure of the full initial Tomita domain, not only its finite-matrix restriction. If $k = A\Omega_b$ with bounded A, then

$$\|\Delta_b^{1/2}k\|_2^2 = \sum_{m,n} p_m|A_{mn}|^2 = \mathrm{Tr}(\rho_bAA^*) \leq \|A\|^2.$$

Hence every such k is in that weighted domain, and the coefficient formula gives $J\Delta_b^{1/2}A\Omega_b = A^*\Omega_b$. The closed weighted operator thus extends the full initial Tomita operator. Conversely its finite-matrix core belongs to that initial domain, proving equality of their closures. The formulas prove $S_b = J\Delta_b^{1/2}$ after closure, and give the modular group with exactly the Connes–Rovelli Section 2.1 sign convention [5]:

$$\alpha_s^{(b)}(A) = \Delta_b^{-is} A \Delta_b^{is} = \rho_b^{-is} A \rho_b^{is} = e^{ibsL} A e^{-ibsL} = \sigma_{bs}(A), \qquad (\alpha_s^{(b)})^{-1} = \alpha_{-s}^{(b)}. \tag{5.8}$$

The middle identities are identities of left multiplication operators on $\mathcal{K}$, as is checked on every E_{mn}. The source deliberately reverses the common mathematical modular sign. Its KMS strip in parameter s has height one. On the algebra of finite words in the generators the modular derivation is

$$\delta_b(A) = ib[L, A], \qquad \delta_b(\mu_n) = ib \log n\, \mu_n, \qquad \delta_b(v(r)) = 0.$$

For these words the commutators are bounded; no assertion that $[L, A]$ is bounded for every $A \in B(\mathcal{H})$ is made.

Retain an energy scale $E_0 > 0$, Boltzmann's constant $k_{\mathrm{B}} > 0$, and Planck's constant $\hbar > 0$. The physical Hamiltonian is $H = E_0 L$, with physical inverse temperature and time evolution

$$\beta_{\mathrm{phys}} = \frac{1}{k_{\mathrm{B}} T} = \frac{b}{E_0}, \qquad \gamma_\theta(A) = e^{i\theta H/\hbar} A e^{-i\theta H/\hbar} = \sigma_{E_0\theta/\hbar}(A).$$

The exact time map and inverse are

$$\alpha_s^{(b)} = \gamma_{\hbar b s/E_0}, \qquad \theta = \frac{\hbar b}{E_0} s = \hbar \beta_{\mathrm{phys}} s, \qquad s = \frac{E_0}{\hbar b}\theta = \frac{k_{\mathrm{B}} T}{\hbar}\theta. \tag{5.9}$$

The physical KMS strip thus has height $\hbar\beta_{\mathrm{phys}}$. This restores the $\hbar$ which Connes–Rovelli set to one immediately after their physical Heisenberg-flow formula in Section 3.1. Equation (5.9) concerns each fixed b; a varying state requires the additional derivative terms calculated below.

5.5 An invertible thermometric image of the actual concentration

The following specifies an auxiliary thermometer completely. Choose the energy scale $E_0 > 0$ already retained above and a calibration inertia $\mathcal{I} > 0$. Let $t_f > 0$ denote the physical time represented by one unit of the original source time. This is a declared unit conversion, with inverse $\theta_{\mathrm{mat}} = t_f t,\ t = \theta_{\mathrm{mat}}/t_f$, not a change of the source equation or of its terminal value $t = 1$. Its measured angular frequency is $|\omega_\nu|/t_f$. Assign the positive energy observable

$$\varepsilon(t) = \frac{\mathcal{I}}{2t_f^2}|\omega_\nu(0, t)|^2 = \frac{2\mathcal{I}}{C^2 t_f^2}(1 - t)^{-2-2h} \qquad (1 - \tau_1 < t < 1). \tag{5.10}$$

Thus $\mathcal{I}$ has the units of moment of inertia and ε has energy units. The vorticity and all original source constants have been retained. The thermometer's calibration rule is that its Gibbs mean energy equal this particular observable.

We prove that the rule determines a unique state, with an explicit inverse coordinate. Define the real functions for $b > 1$

$$\ell(b) = \frac{\sum_{n\ge1} (\log n) n^{-b}}{Z(b)} = -\frac{\zeta'(b)}{\zeta(b)}, \qquad V(b) = \frac{\sum_{n\ge1} (\log n)^2 n^{-b}}{Z(b)} - \ell(b)^2.$$

Every derivative series converges uniformly on compact subsets of $b > 1$, by comparison with $\sum (\log n)^k n^{-1-\delta}$ for a fixed $\delta > 0$. Termwise differentiation therefore gives

$$\ell'(b) = -V(b), \qquad V(b) = \sum_{n\ge1} p_n (\log n - \ell(b))^2 > 0.$$

Strictness follows already from the positive weights of $n = 1, 2$, whose logarithms differ. Consequently the mean energy $U(b) = \phi_b(H) = E_0 \ell(b)$ is smooth and strictly decreasing. As $b \to \infty$, dominated convergence in the two defining series gives $U(b) \to 0$.

Here are explicit estimates at its other endpoint. Put $\delta = b - 1$. For $k = 0, 1, 2$, let $g_k(x) = (\log x)^k x^{-1-\delta}$ on $x \geq 1$. Comparing the integral on every unit interval with its left endpoint gives

$$\left| \sum_{n \geq 1} g_k(n) - \int_1^\infty g_k(x)\, dx \right| \leq \int_1^\infty |g_k'(x)|\, dx.$$

For $k = 0$ the right side is one. For $k \geq 1$, g_k rises from zero to its single maximum and returns to zero, so it equals $2[k/(e(1+\delta))]^k$, which is uniformly bounded for $0 < \delta \leq 1$. Substitution $y = \log x$, followed by k integrations by parts, gives the exact integral $\int_0^\infty y^k e^{-\delta y} dy = k!/\delta^{k+1}$. It follows that

$$Z(1+\delta) = \delta^{-1} + O(1), \quad \ell(1+\delta) = \delta^{-1} + O(1), \quad V(1+\delta) = \delta^{-2} + O(\delta^{-1}). \tag{5.11}$$

In particular $U(b) \to \infty$ as $b \downarrow 1$. The intermediate value theorem and strict monotonicity now prove that $U : (1, \infty) \to (0, \infty)$ is bijective. Its derivative never vanishes, so its inverse is smooth.

Set $A_{\mathcal{I}} = 2\mathcal{I}/(C^2 t_f^2)$ and $a = 2 + 2h$, retaining these full defining constants in what follows. The actual thermometric path and its inverse are

$$b(t) = U^{-1}\left(A_{\mathcal{I}}(1-t)^{-a}\right), \qquad t(b) = 1 - \left(\frac{2\mathcal{I}}{C^2 t_f^2 E_0 \ell(b)}\right)^{1/(2+2h)}. \tag{5.12}$$

Their exact domains are

$$t \in (1 - \tau_1, 1), \qquad b \in (1, b_1), \qquad b_1 = U^{-1}(A_{\mathcal{I}} \tau_1^{-a}).$$

Substitution proves both inverse compositions. No part of the axis formula has been extended to earlier times. Differentiation yields the complete pulled-back state parameter:

$$\varepsilon'(t) = \frac{2\mathcal{I}(2+2h)}{C^2 t_f^2}(1-t)^{-3-2h}, \qquad b'(t) = -\frac{\varepsilon'(t)}{E_0 V(b(t))} < 0. \tag{5.13}$$

The temperature, physical inverse temperature, entropy, and heat capacity of this specified thermometer are

$$T(t) = \frac{E_0}{k_{\mathrm{B}} b(t)}, \qquad \beta_{\mathrm{phys}}(t) = \frac{b(t)}{E_0}, \tag{5.14}$$

$$\mathscr{S}(t) = -k_{\mathrm{B}} \operatorname{Tr}(\rho_{b(t)} \log \rho_{b(t)}) = k_{\mathrm{B}}\{\log Z(b(t)) + b(t)\ell(b(t))\}, \qquad \mathscr{C}(b) = k_{\mathrm{B}} b^2 V(b) > 0. \tag{5.15}$$

These equalities follow respectively from (5.7), $\log p_n = -b \log n - \log Z$, and $(dU/db)/(dT/db) = (-E_0 V)/[-E_0/(k_{\mathrm{B}} b^2)]$. All entropy series are finite for each $b > 1$. Differentiation, using $Z'/Z = -\ell$, gives

$$\frac{d\mathscr{S}}{db} = -k_{\mathrm{B}} b V(b), \qquad \frac{d\mathscr{S}}{dt} = \frac{k_{\mathrm{B}} b(t)}{E_0} \varepsilon'(t) = \frac{\varepsilon'(t)}{T(t)} > 0.$$

This is the exact Gibbs first-law differential for the calibrated state path. It is separate from the bath entropy (5.4), which was calculated directly from viscous stress over the complete fluid support.

The terminal behavior can also be computed. Writing $\tau = 1 - t$, (5.11) and $E_0 \ell(b(t)) = A_{\mathcal{I}} \tau^{-a}$ prove

$$b(t) - 1 = \frac{E_0}{A_{\mathcal{I}}} \tau^a (1 + o(1)), \qquad b'(t) = -\frac{a E_0}{A_{\mathcal{I}}} \tau^{a-1}(1 + o(1)), \tag{5.16}$$

$$T(t) \longrightarrow E_0/k_{\mathrm{B}} \quad \text{from below}, \qquad \mathscr{C}(b(t)) = k_{\mathrm{B}} \left(\frac{\varepsilon(t)}{E_0}\right)^2 (1 + o(1)). \tag{5.17}$$

Finally $\delta\ell(1+\delta) \to 1$ and $Z(1+\delta)/\ell(1+\delta) \to 1$. Substituting these two limits into (5.15) proves the more precise entropy image

$$\mathscr{S}(t) = k_{\mathrm{B}} \left\{ \frac{\varepsilon(t)}{E_0} + \log \frac{\varepsilon(t)}{E_0} + 1 + o(1) \right\}. \tag{5.18}$$

Thus the actual source axis concentration reaches the $b \downarrow 1$ edge of this Gibbs representation, with a finite limiting calibrated temperature and diverging mean energy, entropy, and heat capacity. These conclusions retain the stated calibration constants and do not assign this equation of state to an unspecified physical fluid.

5.6 The terminal state exists on the represented source algebra

Failure of the trace sum at $b = 1$ does not end the correspondence. We calculate its endpoint on $\mathcal{A}$ directly. Finite linear combinations of $\mu_n v(r)\mu_m^*$ form a dense star algebra. Indeed $\mu_m^*\mu_a = \mu_{a/d}\mu_{m/d}^*$, $d = \gcd(m,a)$, as is checked on every basis vector. The product of two displayed monomials is consequently

$$[\mu_n v(r)\mu_m^*][\mu_a v(s)\mu_b^*] = \mu_{na/d}\, v((a/d)r + (m/d)s)\, \mu_{bm/d}^*.$$

Adjoints have the same form and the generators belong to this algebra, proving density.

Taking diagonal matrix entries in (5.7) gives the exact formula

$$\phi_b(\mu_n v(r)\mu_m^*) = \begin{cases} 0, & n \neq m, \\ n^{-b} Z(b)^{-1} \sum_{j \geq 1} e^{2\pi i j r} j^{-b}, & n = m. \end{cases}$$

When $r = 0$ the sum is $Z(b)$. When $r \neq 0$ in $\mathbb{Q}/\mathbb{Z}$, the partial sums of $e^{2\pi i j r}$ are bounded by $2/|1 - e^{2\pi i r}|$. Summation by parts bounds the whole series uniformly for $1 \leq b \leq 2$, since $\int_1^\infty b x^{-b-1} dx = 1$. Therefore

$$\lim_{t \uparrow 1} \phi_{b(t)}(\mu_n v(r)\mu_m^*) = \phi_{\mathrm{c}}(\mu_n v(r)\mu_m^*) = \mathbf{1}_{n=m}\, \mathbf{1}_{r=0}\, \frac{1}{n}. \tag{5.19}$$

The limit extends to every $A \in \mathcal{A}$: approximate in norm by a finite linear combination of monomials, and bound the two state errors by that norm. Positivity and value one at the identity pass to the limit. This constructs the state ϕ_{c}, together with the exact state-space map from the actual preterminal path.

It is a 1-KMS state for the same σ. On the dense monomial algebra each element is a finite sum of eigenoperators of σ, hence entire analytic. The b-KMS identity $\phi_b(B\sigma_{ib}(A)) = \phi_b(AB)$ passes to the limit: $\sigma_{ib}(A) \to \sigma_i(A)$ in norm there. Equivalently, the finite exponential functions $\phi_b(B\sigma_{bz}(A))$ on the unit-height strip converge to functions with the required boundary values. Norm approximation and the same strip maximum estimate used above extend these functions to arbitrary $A, B \in \mathcal{A}$.

This endpoint has no normal extension to $B(\mathcal{H})$ in the fixed representation. For otherwise a density operator D would have diagonal probabilities $q_m = \langle e_m, De_m\rangle \geq 0$, $\sum q_m = 1$. The diagonal operators $v(1/k)$ then satisfy

$$\mathrm{Tr}(Dv(1/k)) = \sum_{m \geq 1} q_m e^{2\pi i m/k} \longrightarrow 1 \quad (k \to \infty)$$

by dominated convergence. But (5.19) assigns zero to each $v(1/k)$ for $k \geq 2$, a contradiction. We have therefore computed both the terminal state on the source algebra and the exact obstruction to retaining its preterminal normal density representation.

5.7 The actual state derivative and all modular clock terms

For every fixed bounded A, differentiation of the absolutely convergent diagonal trace on a compact preterminal interval gives

$$\frac{d}{dt}\phi_{b(t)}(A) = -b'(t)\left\{\mathrm{Tr}(\rho_{b(t)}LA) - \ell(b(t))\,\mathrm{Tr}(\rho_{b(t)}A)\right\}. \tag{5.20}$$

Here $\rho_b L$ is trace class, so the expression is defined for all bounded A, whether or not A preserves $D(L)$. The eigenvalue calculation behind it is

$$\frac{dp_n(t)}{dt} = -b'(t)(\log n - \ell(b(t)))p_n(t).$$

In particular the fixed bounded source-algebra projection $P_n = \mu_n\mu_n^*$ has

$$\phi_{b(t)}(P_n) = n^{-b(t)}, \qquad \frac{d}{dt}\phi_{b(t)}(P_n) = -b'(t)\log n\, n^{-b(t)} > 0 \quad (n \geq 2).$$

This explicitly detects the changing state using an observable already in $\mathcal{A}$, rather than only an unbounded energy.

At each fixed t its modular generator is

$$\delta_{b(t)}(A) = ib(t)[L, A], \qquad \delta_{b(t)}(\mu_n) = i\log n\, U^{-1}\left(\frac{2\mathcal{I}}{C^2t_f^2}(1-t)^{-2-2h}\right)\mu_n. \tag{5.21}$$

This is the actual pullback through (5.12), on the finite-word domain specified above. For each such word the coefficient tends to the finite critical value as $t \uparrow 1$, although the state's mean energy has the divergence (5.10). The changing state cannot be its own stationary modular evolution: each modular group fixes its state, whereas the displayed expectation of P_n changes. More strongly, two distinct ρ_b cannot be related by any unitary conjugation on $\mathcal{H}$. Their unique largest eigenvalues are $Z(b)^{-1}$, strictly increasing in b; unitary conjugation preserves the largest eigenvalue.

There are distinct time derivatives which should not be conflated. For a fixed modular parameter s, the explicit exponential in (5.8) gives

$$\partial_t\alpha_s^{(b(t))}(A) = isb'(t)[L, \alpha_s^{(b(t))}(A)].$$

For any differentiable modular path $s(t)$, the full chain rule is

$$\frac{d}{dt}\alpha_{s(t)}^{(b(t))}(A) = i\{b(t)s'(t) + s(t)b'(t)\}[L, \alpha_{s(t)}^{(b(t))}(A)]. \tag{5.22}$$

Both are identities on finite words; the state derivative (5.20) is an additional term when an expectation of a varying observable is differentiated.

Fix $t_0 \in (1-\tau_1, 1)$ and use $t \in [t_0, 1)$ for the two clocks below. One exact physical-time coordinate is obtained by the pointwise fixed-state dictionary, now used with its complete chain rule:

$$s_{\mathrm{pt}}(t) = \frac{E_0t_f}{\hbar b(t)}(t-t_0), \qquad \alpha_{s_{\mathrm{pt}}(t)}^{(b(t))} = \gamma_{t_f(t-t_0)}.$$

Its derivative satisfies $bs_{\mathrm{pt}}' + s_{\mathrm{pt}}b' = E_0t_f/\hbar$, so (5.22) exactly recovers the fixed Hamiltonian generator $iE_0t_f[L,\cdot]/\hbar$. This equality compares the thermometer's flows; it does not map them to the material flow ruled out in Section 5.3. Moreover $s_{\mathrm{pt}}'(t) > 0$ on this interval, since $b > 0,\ b' < 0,\ t - t_0 \geq 0$. Its inverse is the unique root of $E_0t_f(t-t_0) = \hbar b(t)s_{\mathrm{pt}}$; equivalently substitute the exact $t(b)$ of (5.12) into $s_{\mathrm{pt}} = E_0t_f[t(b) - t_0]/(\hbar b)$, solve its strictly monotone scalar equation for b, and apply $t(b)$.

Alternatively, an accumulated local thermal clock is the explicitly invertible coordinate

$$s_{\rm loc}(t) = \frac{E_0 t_f}{\hbar} \int_{t_0}^{t} \frac{dv}{b(v)}, \qquad \frac{dt}{ds_{\rm loc}} = \frac{\hbar b(t)}{E_0 t_f}.$$

It is strictly increasing; its inverse is the unique root of the displayed increasing integral on the exact preterminal interval. For a nonconstant b, evaluating a current-state modular flow at $s_{\rm loc}(t)$ has the extra coefficient $s_{\rm loc}(t)b'(t)$ in (5.22); dropping it would not reproduce the physical Heisenberg flow. Both clocks have finite preterminal limits as $t \uparrow 1$, because $1 < b(t) < b_1$ and $b(t) \to 1$. These explicit coordinate maps, their inverses, and their full derivatives retain the distinction between a stationary Gibbs dynamics and the actually varying state produced by the source concentration.

References

[1] *Finite Time Blowup for Navier–Stokes*, released 166-page manuscript, source edition identified by SHA-256 in Section 1. https://cdn.openai.com/pdf/32d9f210-8b73-45e0-91bc-82a30aef8a9a/navier-stokes.pdf.

[2] I. Bredberg, C. Keeler, V. Lysov, and A. Strominger, *From Navier–Stokes to Einstein*, arXiv:1101.2451v2. https://arxiv.org/abs/1101.2451v2.

[3] C. Eling, I. Fouxon, and Y. Oz, *The Incompressible Navier–Stokes Equations From Black Hole Membrane Dynamics*, arXiv:0905.3638v2. https://arxiv.org/abs/0905.3638v2.

[4] I. Bredberg and A. Strominger, *Black Holes as Incompressible Fluids on the Sphere*, arXiv:1106.3084v1. https://arxiv.org/abs/1106.3084v1.

[5] A. Connes and C. Rovelli, *Von Neumann Algebra Automorphisms and Time-Thermodynamics Relation in General Covariant Quantum Theories*, arXiv:gr-qc/9406019. https://arxiv.org/abs/gr-qc/9406019.

[6] J.-B. Bost and A. Connes, *Hecke algebras, type III factors and phase transitions with spontaneous symmetry breaking in number theory*, Selecta Mathematica **1** (1995), 411–457. https://doi.org/10.1007/BF01589495.